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PRACTICAL - 10

AIM: Data Preparation and Transformation using Tableau Prep

- a) Connect and load data into Tableau Prep
- b) Clean and profile data
- c) Group and standardize values
- d) Merge mismatched fields
- e) Perform aggregations in Prep
- f) Apply filters and transformations
- g) Create and automate data preparation flows

A: Connect and Load Data into Tableau Prep

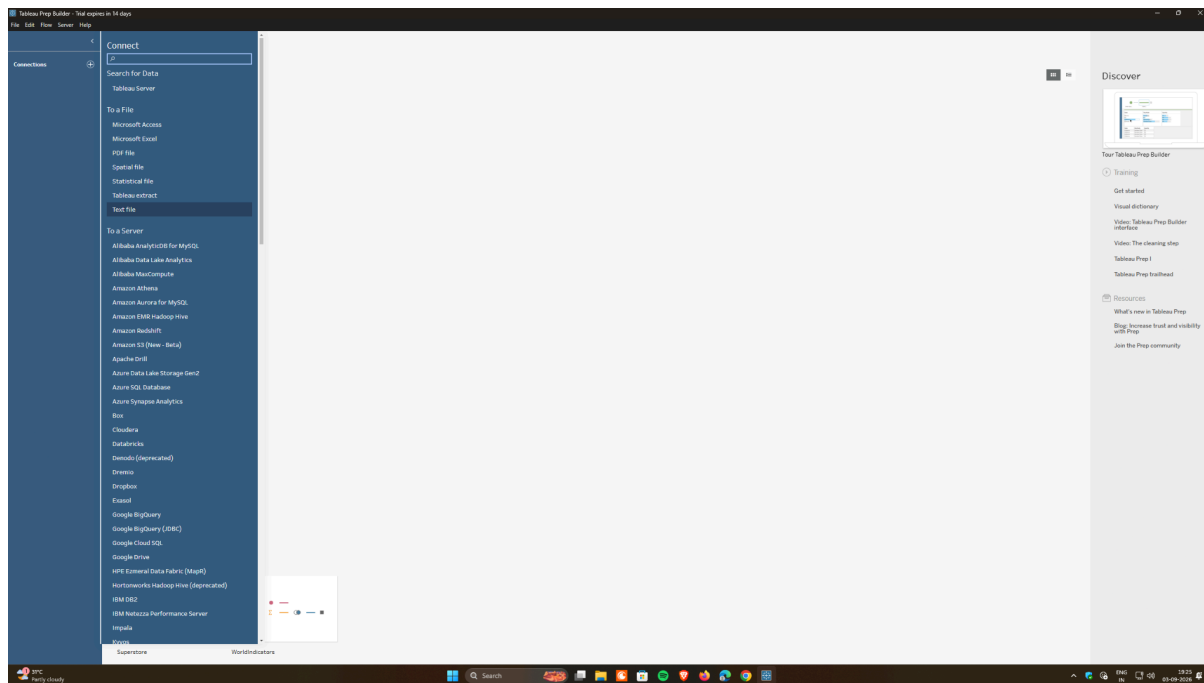
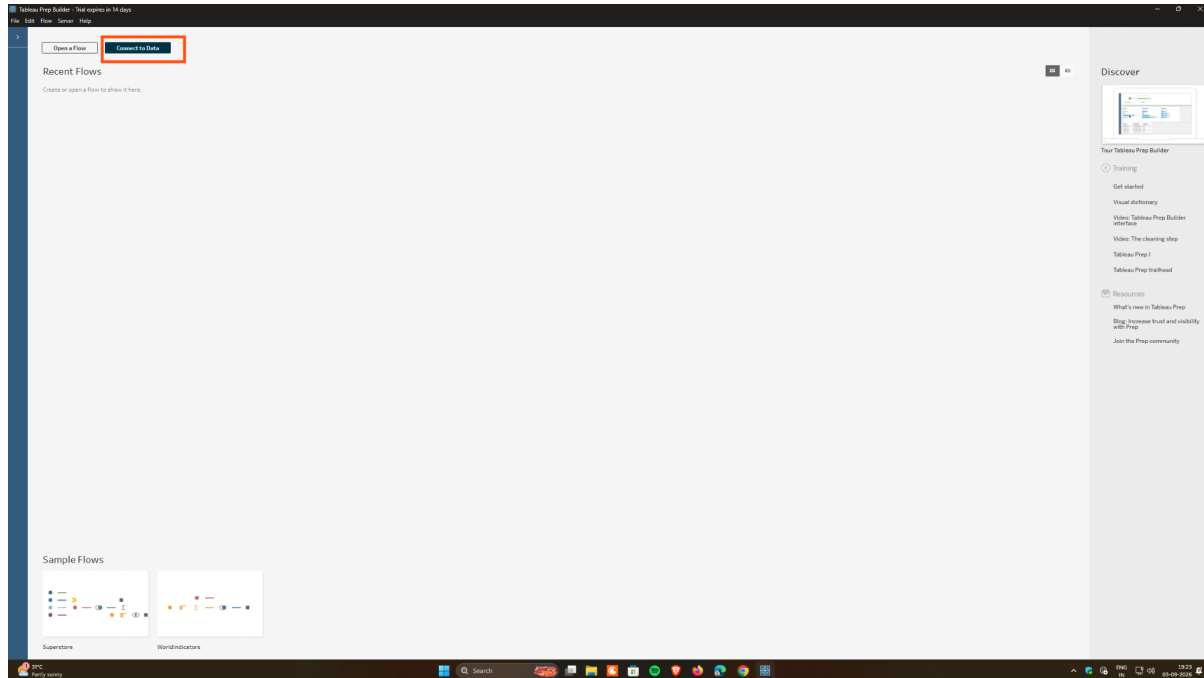
Objective: Establish connections to the local CSV data sources (Student_Assessments_Term1.csv, Student_Assessments_Term2.csv, and Course_Credit_Master.csv) and initiate a visual data preparation workflow.

Tableau Prep Builder utilizes an input step pipeline architecture. Connecting multiple disparate files into a unified workspace establishes the foundation for schema validation, data type detection, and downstream data profiling.

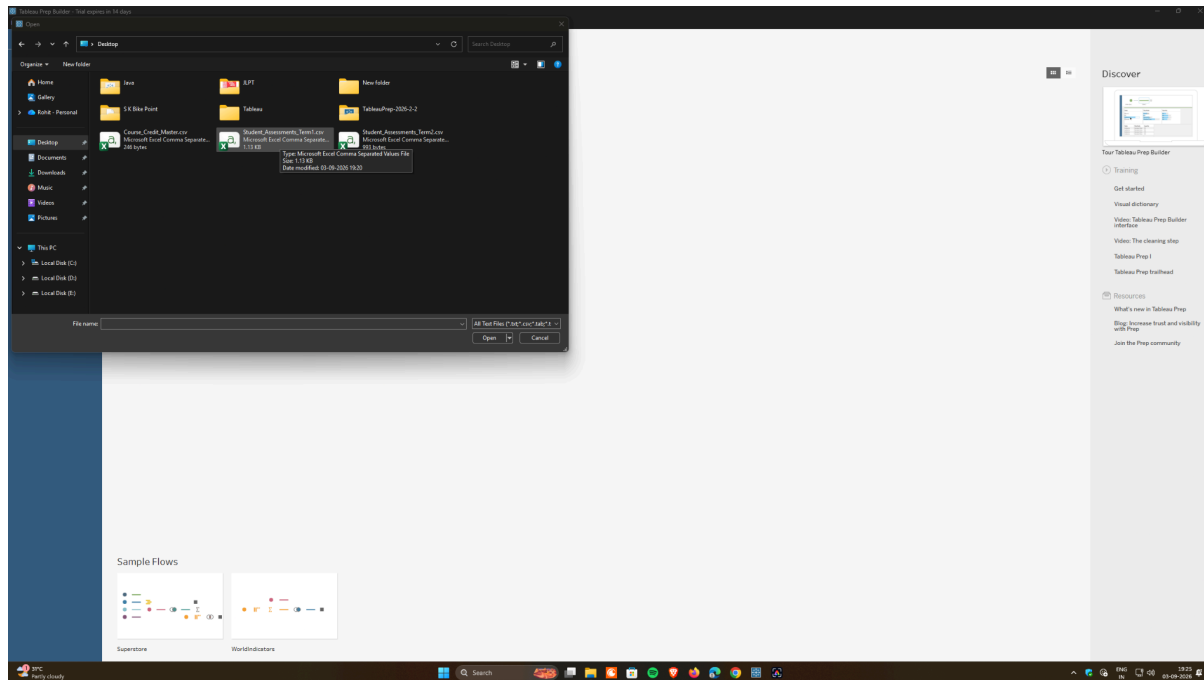
Step 1: Launch **Tableau Prep Builder** and locate the **Connections** pane on the left side of the start screen.

Step 2: In the **Connections** pane, click the + (**Add Connection**) button and select **Student_Assessments_Term1.csv**.

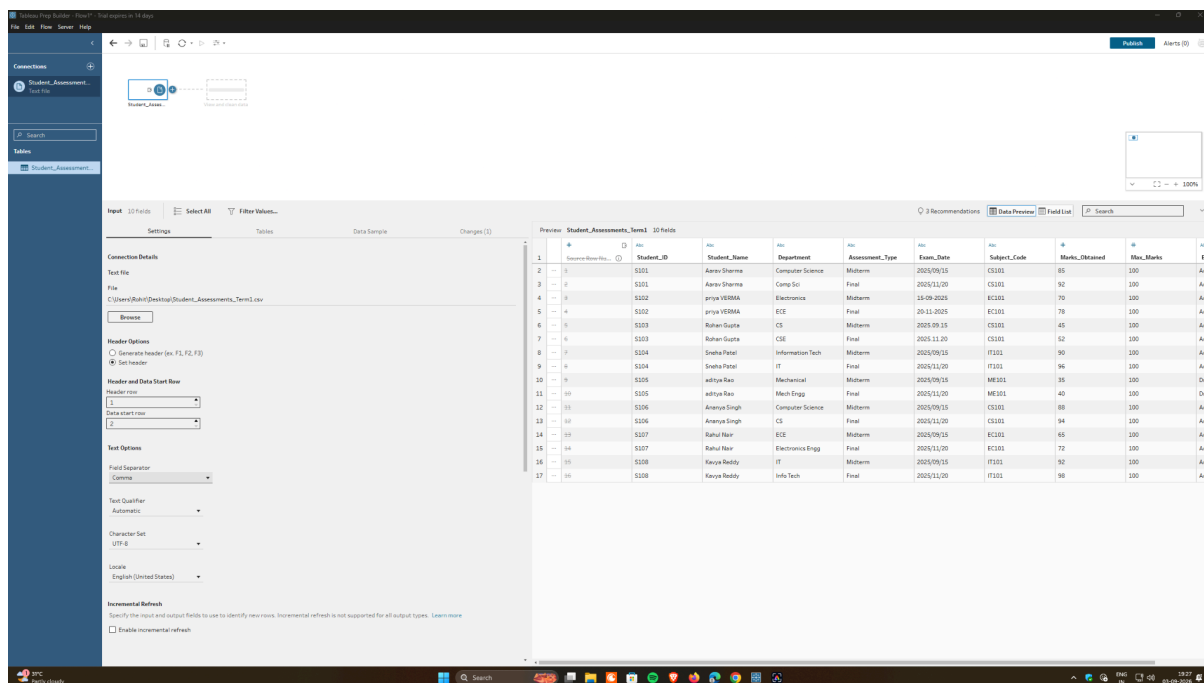
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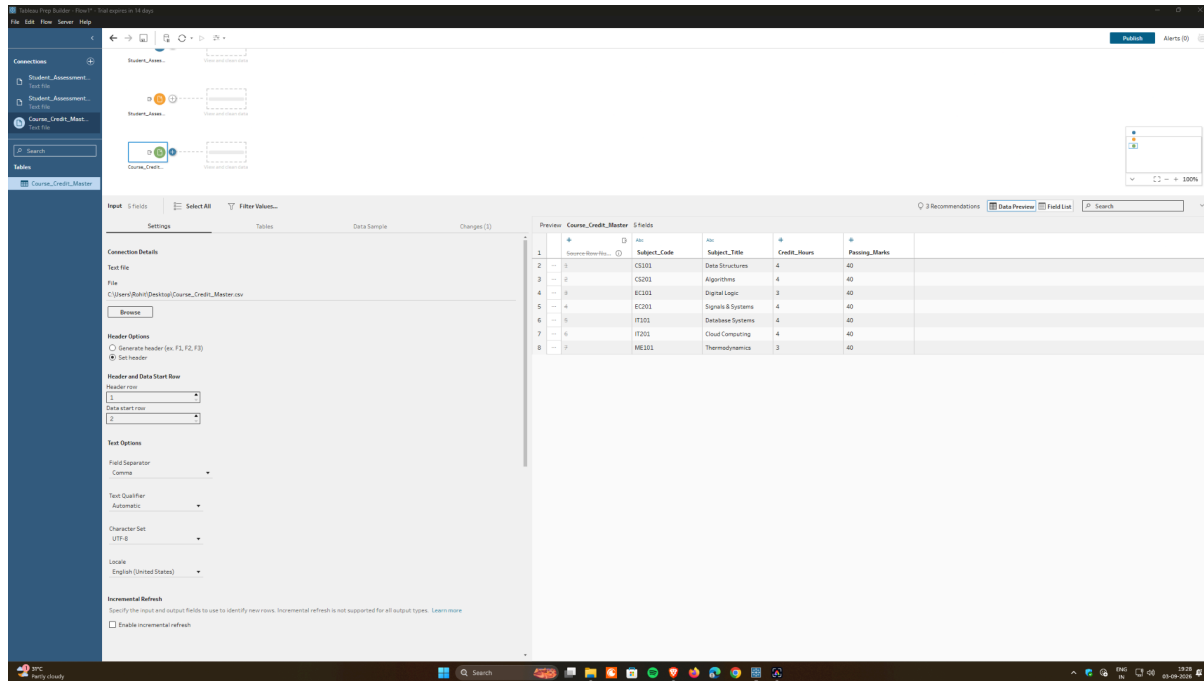


Select **Text file**, browse to **Student_Assessments_Term1.csv**, click **Open**, then verify the **Student_Assessments_Term1** Input Step appears and **Field Separator** is set to **Comma (,)**.



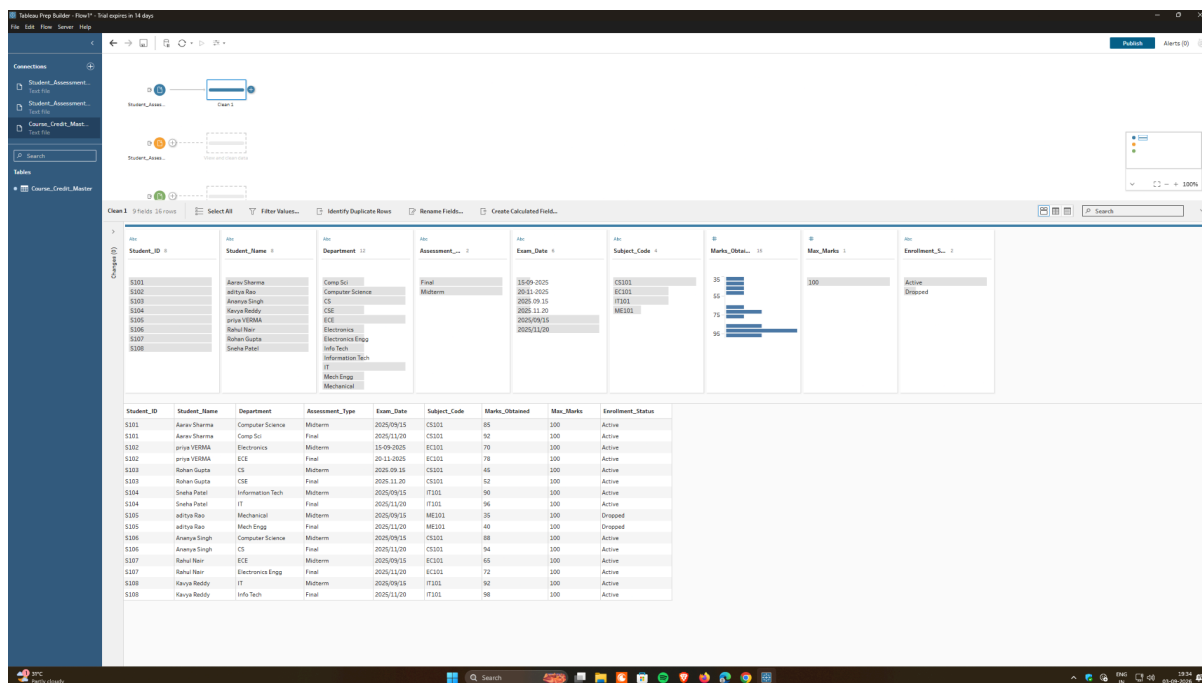
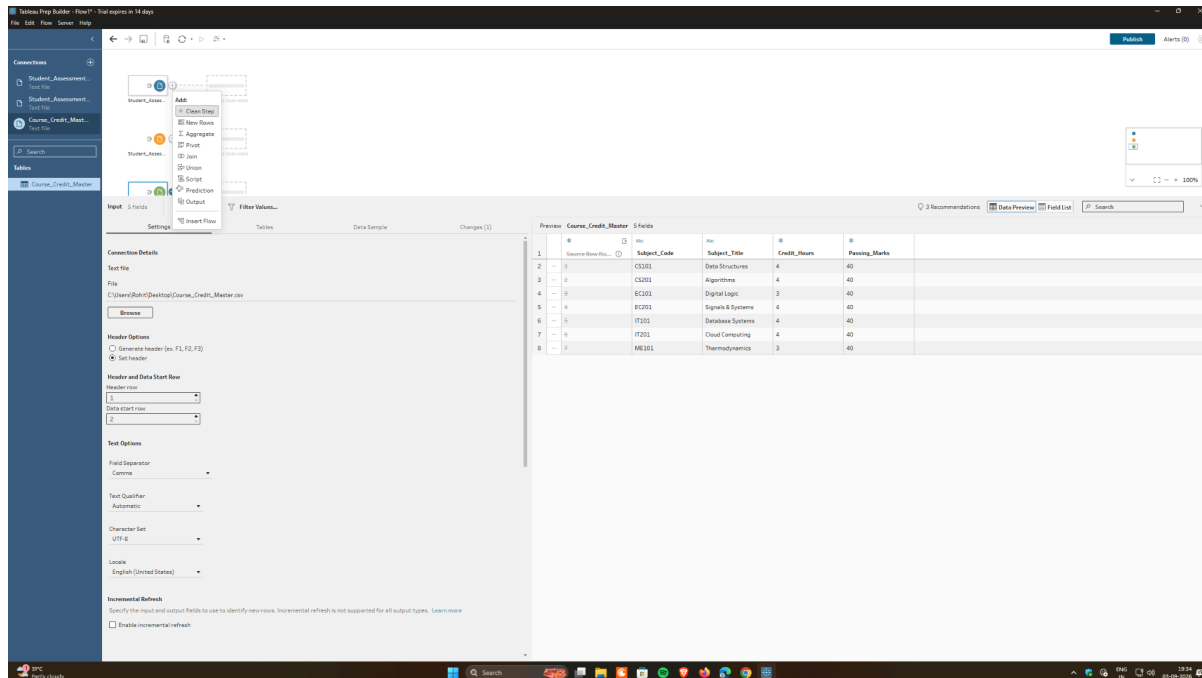
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Step 3: Click + (Add Connection) twice, select **Text file** each time, and open **Student_Assessments_Term2.csv** and **Course_Credit_Master.csv**, resulting in three separate **Input** icons on the Flow canvas.



Step 4: Hover over **Student_Assessments_Term1**, click the + icon, select **Clean Step**, and verify that the **Profile Pane** opens showing cards for all 9 columns.

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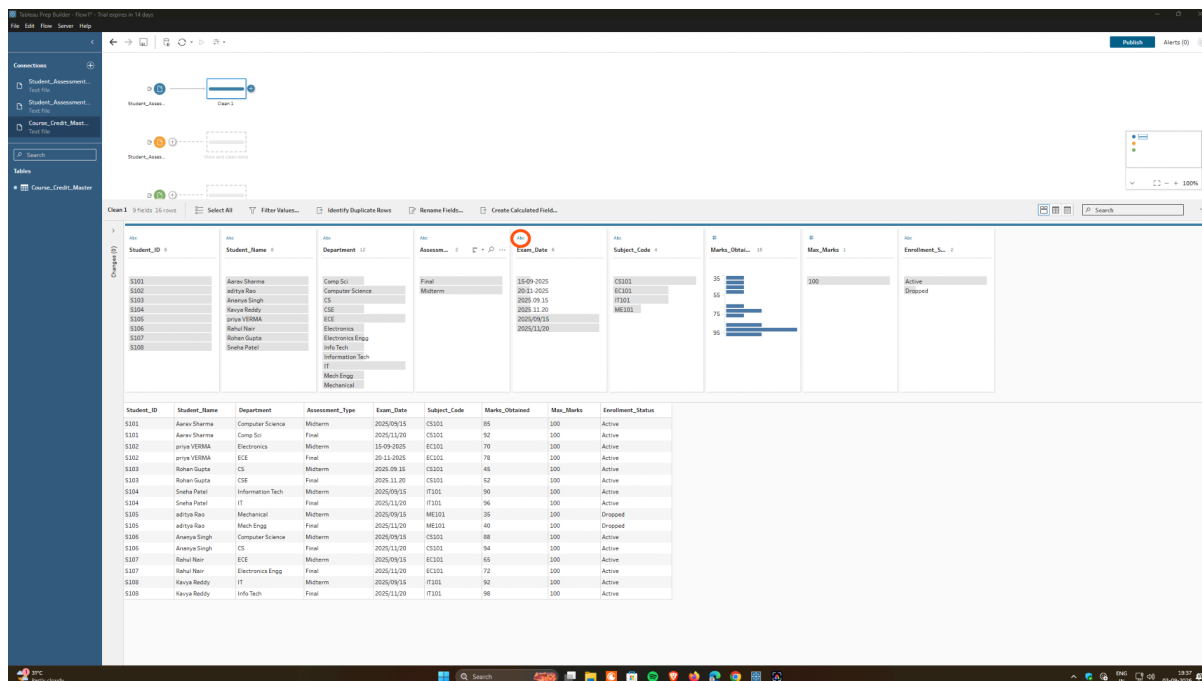
B: Clean and Profile Data

Objective: Use Tableau Prep Builder's interactive Profile Pane to identify data anomalies, whitespace issues, date formatting discrepancies, and inconsistent casing, then apply standard cleaning operations.

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Unlike traditional spreadsheets, Tableau Prep provides real-time profiling cards that visualize distributions, frequency counts, and metadata data types for every field. Profiling allows you to quickly detect hidden formatting defects (such as extra spaces in names like ' Aarav Sharma ' or non-standard date strings) before standardizing categorical groups or performing joins.

Step 1: Select the **Clean 1** step and inspect the **Profile Pane** to identify inconsistent **Student_Name** values (extra spaces/mixed casing) and **Exam_Date** stored as a string due to mixed separators (/, -, .).



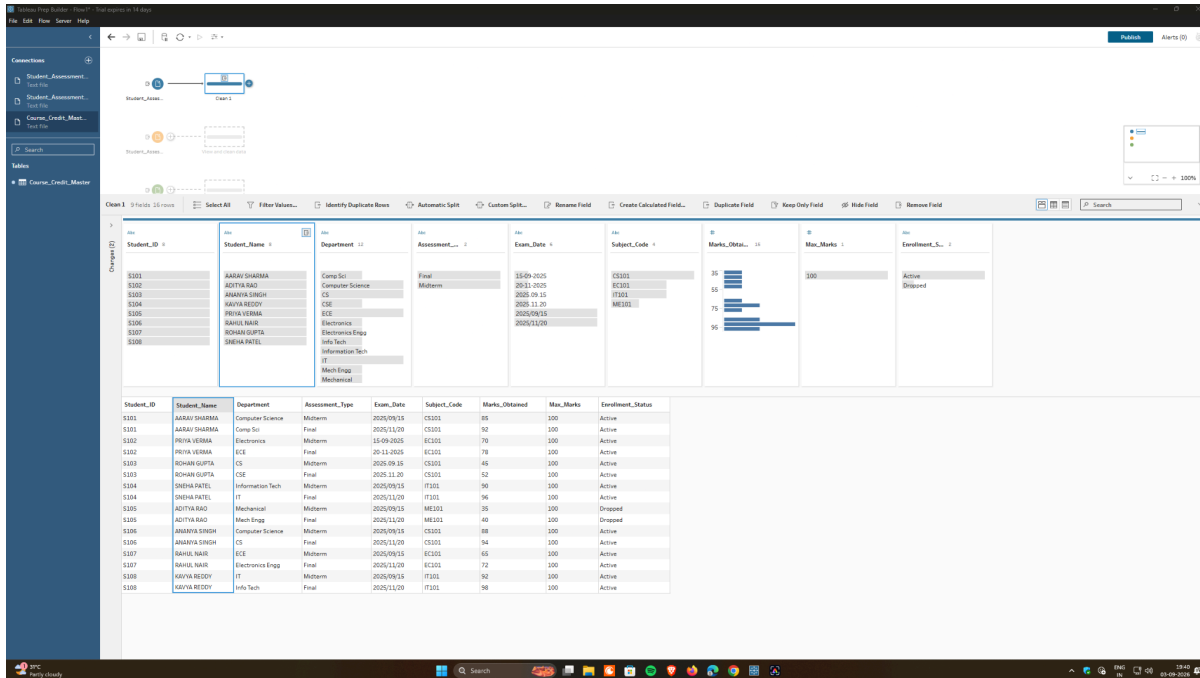
Step 2: On the **Student_Name** card, select **More options (...)** → **Clean** → **Trim Spaces**, then repeat with **Make Uppercase** to remove extra spaces and standardize the names.

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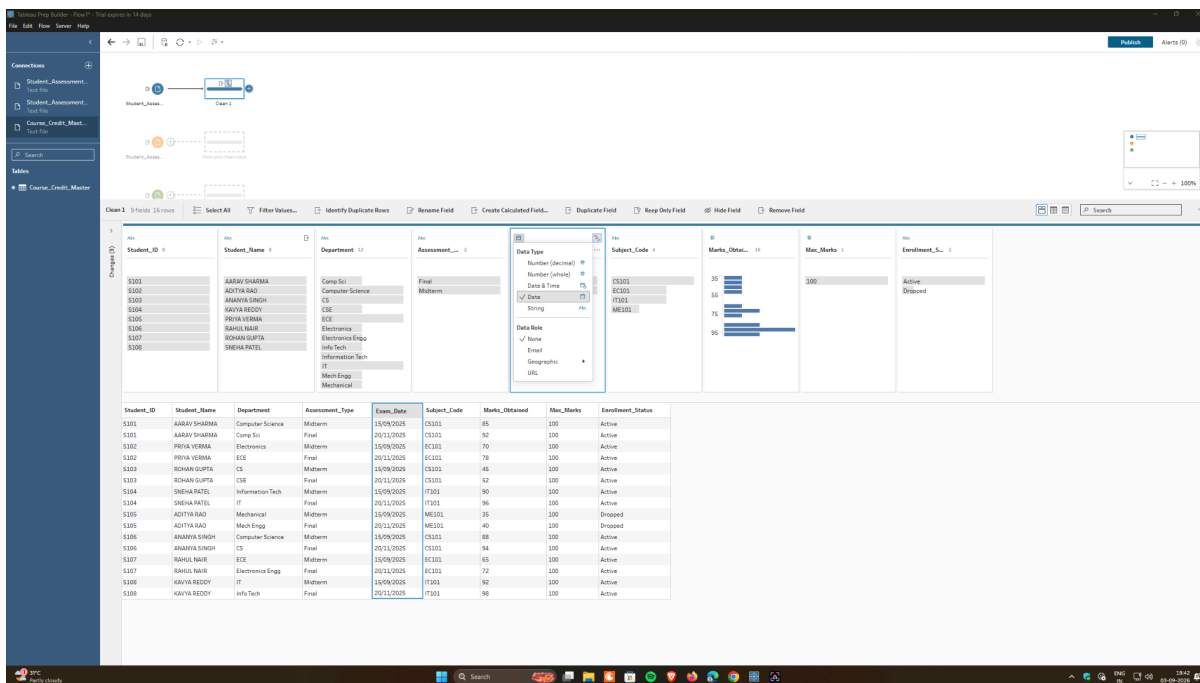
Student_ID	Student_Name	Department	Ex_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S1001	Aarav Sharma	Computer Science	10/15	CS101	85	100	Active
S1001	Aarav Sharma	Comp Sci	11/20	CS101	92	100	Active
S1002	Aranya VERMA	Electronics	15/09-2025	EC101	70	100	Active
S1002	Aranya VERMA	ECE	20-11-2025	EC101	78	100	Active
S1003	Rohan Gupta	CS	2025-09-15	CS101	45	100	Active
S1003	Rohan Gupta	CSE	2025-11-20	CS101	52	100	Active
S1004	Snaha Patel	Information Tech	2025/09/15	IT101	90	100	Active
S1004	Snaha Patel	IT	2025/11/20	IT101	96	100	Active
S1005	Aditya Rao	Mechanical	2025/09/15	ME101	35	100	Dropped
S1005	Aditya Rao	Mech Engg	2025/11/20	ME101	40	100	Dropped
S1006	Ananya Singh	Computer Science	2025/09/15	CS101	88	100	Active
S1006	Ananya Singh	CS	2025/11/20	CS101	94	100	Active
S1007	Rahul Nair	ECE	2025/09/15	EC101	65	100	Active
S1007	Rahul Nair	Electronics Engg	2025/11/20	EC101	72	100	Active
S1008	Kavya Reddy	IT	2025/09/15	IT101	92	100	Active
S1008	Kavya Reddy	Info Tech	2025/11/20	IT101	98	100	Active

Student_ID	Student_Name	Department	Ex_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S1001	Aarav Sharma	Computer Science	10/15	CS101	85	100	Active
S1001	Aarav Sharma	Comp Sci	11/20	CS101	92	100	Active
S1002	Aranya VERMA	Electronics	15/09-2025	EC101	70	100	Active
S1002	Aranya VERMA	ECE	20-11-2025	EC101	78	100	Active
S1003	Rohan Gupta	CS	2025-09-15	CS101	45	100	Active
S1003	Rohan Gupta	CSE	2025-11-20	CS101	52	100	Active
S1004	Snaha Patel	Information Tech	2025/09/15	IT101	90	100	Active
S1004	Snaha Patel	IT	2025/11/20	IT101	96	100	Active
S1005	Aditya Rao	Mechanical	2025/09/15	ME101	35	100	Dropped
S1005	Aditya Rao	Mech Engg	2025/11/20	ME101	40	100	Dropped
S1006	Ananya Singh	Computer Science	2025/09/15	CS101	88	100	Active
S1006	Ananya Singh	CS	2025/11/20	CS101	94	100	Active
S1007	Rahul Nair	ECE	2025/09/15	EC101	65	100	Active
S1007	Rahul Nair	Electronics Engg	2025/11/20	EC101	72	100	Active
S1008	Kavya Reddy	IT	2025/09/15	IT101	92	100	Active
S1008	Kavya Reddy	Info Tech	2025/11/20	IT101	98	100	Active

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Step 3: On the **Exam_Date** card, click the **Abc data type icon**, select **Date**, and let Tableau Prep automatically standardize the mixed date formats into calendar dates.



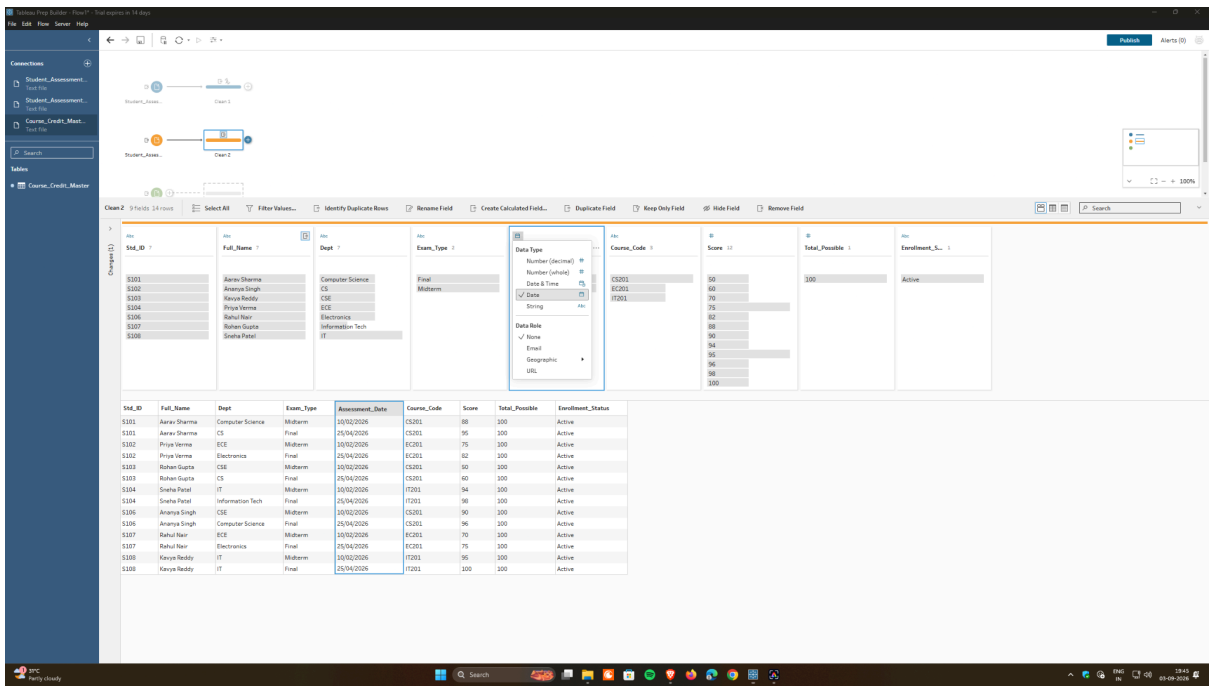
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Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S1001	AARAV SHARMA	Computer Science	Midterm	15/09/2025	CS101	85	100	Active
S1001	AARAV SHARMA	Comp Sci	Final	20/11/2025	CS101	92	100	Active
S1002	PRIVA VERMA	Electronics	Midterm	15/09/2025	EC101	70	100	Active
S1002	PRIVA VERMA	ECE	Final	20/11/2025	EC101	78	100	Active
S1003	ROHAN GUPTA	CS	Midterm	15/09/2025	CS101	45	100	Active
S1003	ROHAN GUPTA	CS	Final	20/11/2025	CS101	52	100	Active
S1004	SNEHA PATEL	Information Tech	Midterm	15/09/2025	IT101	90	100	Active
S1004	SNEHA PATEL	IT	Final	20/11/2025	IT101	96	100	Active
S1005	ADITYA RAO	Mechanical	Midterm	15/09/2025	ME101	35	100	Dropped
S1005	ADITYA RAO	Mech Engg	Final	20/11/2025	ME101	40	100	Dropped
S1006	ANANYA SINGH	Computer Science	Midterm	15/09/2025	CS101	88	100	Active
S1006	ANANYA SINGH	CS	Final	20/11/2025	CS101	94	100	Active
S1007	RAHUL NAR	ECE	Midterm	15/09/2025	EC101	65	100	Active
S1007	RAHUL NAR	Electronics Engg	Final	20/11/2025	EC101	72	100	Active
S1008	KAVYA REDDY	IT	Midterm	15/09/2025	IT101	92	100	Active
S1008	KAVYA REDDY	Info Tech	Final	20/11/2025	IT101	98	100	Active

Step 4: Add a **Clean Step** to **Student_Assessments_Term2**, then in **Clean 2** apply **Trim Spaces** to **Full_Name** and change **Assessment_Date** from **Abc** to **Date**.

Student_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S1001	Aarav Sharma	Computer Science	Final	20/11/2025	CS101	92	100	Active
S1001	Aarav Sharma	CS	Midterm	15/09/2025	CS101	85	100	Active
S1002	Priva Verma	ECE	Final	20/11/2025	EC101	78	100	Active
S1002	Priva Verma	Electronics	Midterm	15/09/2025	EC101	70	100	Active
S1003	Rohan Gupta	CS	Final	20/11/2025	CS101	60	100	Active
S1003	Rohan Gupta	CS	Midterm	15/09/2025	CS101	45	100	Active
S1004	Sneha Patel	IT	Final	20/11/2025	IT101	94	100	Active
S1004	Sneha Patel	Information Tech	Midterm	15/09/2025	IT101	90	100	Active
S1005	Aditya Rao	Mechanical	Final	20/11/2025	ME101	40	100	Dropped
S1005	Aditya Rao	Mech Engg	Midterm	15/09/2025	ME101	35	100	Dropped
S1006	Ananya Singh	Computer Science	Final	20/11/2025	CS101	94	100	Active
S1006	Ananya Singh	CS	Midterm	15/09/2025	CS101	88	100	Active
S1007	Rahul Nar	ECE	Final	20/11/2025	EC101	72	100	Active
S1007	Rahul Nar	Electronics	Midterm	15/09/2025	EC101	65	100	Active
S1008	Kavya Reddy	IT	Final	20/11/2025	IT101	98	100	Active
S1008	Kavya Reddy	Info Tech	Midterm	15/09/2025	IT101	92	100	Active

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Step 5: Open the **Changes** tab on the left to review the itemized, reversible transformations, such as **Trim Spaces** and **data type changes**.

Profile & Cleaning Audit Table:

Field Name	Detected Issue	Cleaning Operation Applied	Target State
Student_Name	Leading/trailing spaces (' Aarav Sharma ')	Clean Trim Spaces	Single trimaligned name
Student_Name	Inconsistent casing ('priya VERMA')	Clean Make Uppercase	Standardized casing
Exam_Date	Text datatype with mixed separators	Change Data Type Date	Standard Date format (YYYYMM-DD)
Assessment_Date	String data type	Change Data Type Date	Standard Date format (YYYYMM-DD)

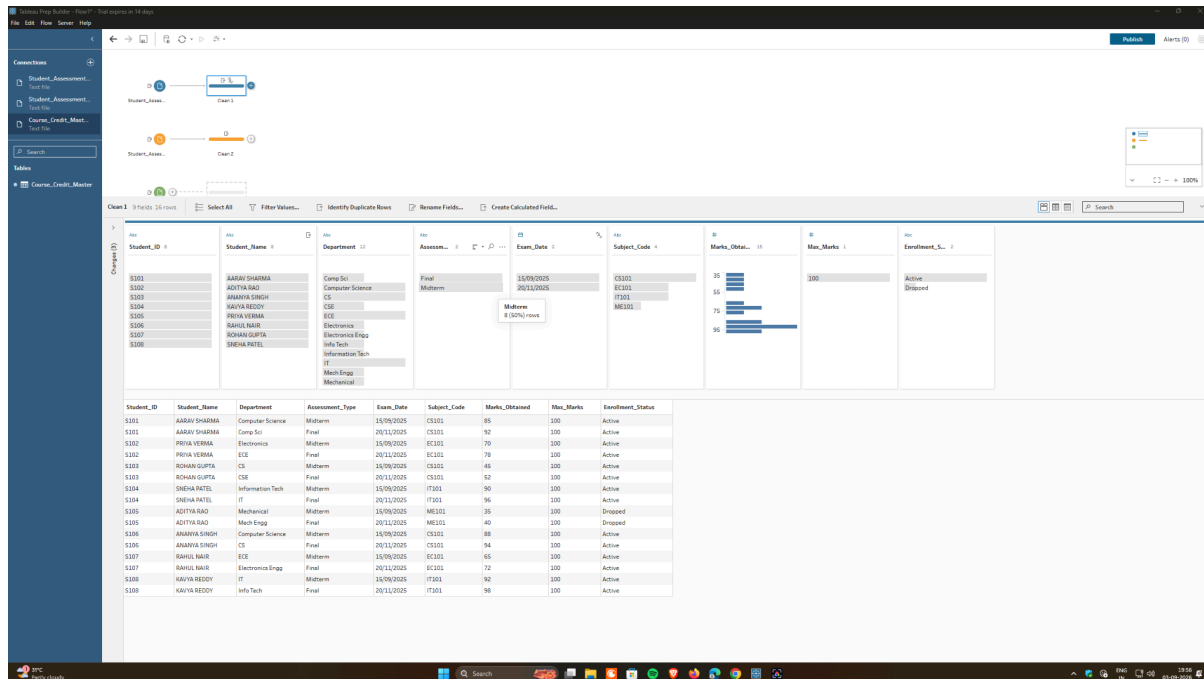
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C: Group and Standardize Values

Objective: Standardize spelling variations, abbreviations, and synonyms in categorical columns (such as Department and Dept) into clean, consistent values using Tableau Prep's built-in grouping algorithms.

In academic datasets, departments are frequently entered under different aliases (e.g., "Comp Sci", "CS", "CSE", and "Computer Science"). If left unstandardized, downstream visual analysis will treat each alias as a completely different academic department, splintering the summary metrics.

Step 1: In **Clean 1**, locate the **Department** card and identify the 12 distinct entries that represent 4 actual departments: **Computer Science, Electronics, Information Technology, and Mechanical Engineering.**



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Step 2: On the **Department** card, select **More options (...)** → **Group Values** → **Manual Selection**, choose **Computer Science**, then group **Comp Sci**, **CS**, and **CSE** with it.

The screenshot shows the Power BI Desktop interface with the 'Department' card selected. The 'More options' menu is open, and the 'Group Values' option is selected. The 'Manual Selection' option is also visible. The 'Department' card is currently showing 'Computer Science' as the selected value. The 'Assessment_Type' card is showing 'Final' as the selected value. The 'Exam_Date' card is showing '15/09/2025' as the selected value. The 'Subject_Code' card is showing 'CS101' as the selected value. The 'Marks_Obtain...' card is showing '35' as the selected value. The 'Max_Marks' card is showing '100' as the selected value. The 'Enrollment_Sta...' card is showing 'Active' as the selected value. The 'Student_ID' card is showing '1301' as the selected value. The 'Student_Name' card is showing 'AARAV SHARMA' as the selected value. The 'Assessment_Type' card is showing 'Final' as the selected value. The 'Exam_Date' card is showing '15/09/2025' as the selected value. The 'Subject_Code' card is showing 'CS101' as the selected value. The 'Marks_Obtain...' card is showing '35' as the selected value. The 'Max_Marks' card is showing '100' as the selected value. The 'Enrollment_Sta...' card is showing 'Active' as the selected value. The 'Student_ID' card is showing '1301' as the selected value. The 'Student_Name' card is showing 'AARAV SHARMA' as the selected value.

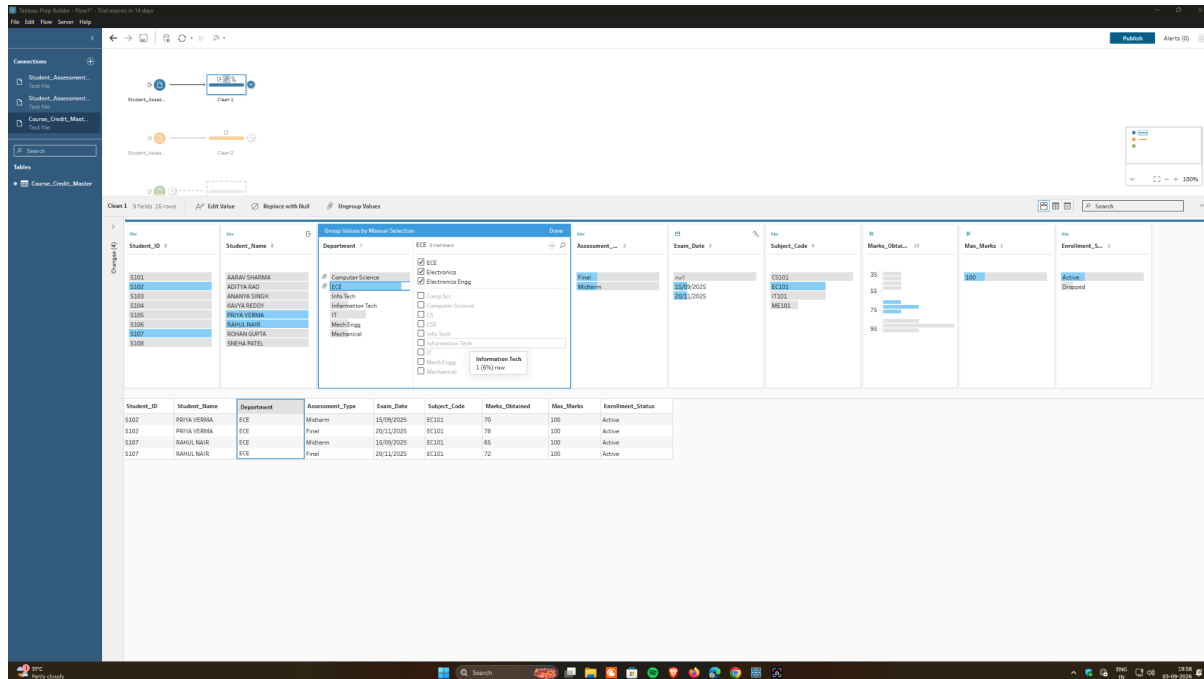
Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtain...	Max_Marks	Enrollment_Sta...
1301	AARAV SHARMA	Computer Science	Midterm	15/09/2025	CS101	35	100	Active
1302	AARAV SHARMA	Comp Sci	Final	20/11/2025	CS101	92	100	Active
1303	PRITHA VERMA	Electronics	Midterm	15/09/2025	EC101	79	100	Active
1304	PRITHA VERMA	ECE	Final	20/11/2025	EC101	78	100	Active
1305	ROHAN GUPTA	CS	Midterm	15/09/2025	CS101	46	100	Active
1306	ANANYA SINGH	CSE	Final	20/11/2025	CS101	94	100	Active
1307	RAHUL NAIR	IT	Midterm	15/09/2025	IT101	92	100	Active
1308	RAHUL NAIR	Information Tech	Final	20/11/2025	IT101	96	100	Active
1309	ADITYA RAO	Mechanical	Midterm	15/09/2025	ME101	35	100	Dropped
1310	ADITYA RAO	Mech Engg	Final	20/11/2025	ME101	49	100	Dropped

The screenshot shows the Power BI Desktop interface with the 'Department' card selected. The 'More options' menu is open, and the 'Group Values' option is selected. The 'Manual Selection' option is also visible. The 'Department' card is currently showing 'Computer Science' as the selected value. The 'Assessment_Type' card is showing 'Final' as the selected value. The 'Exam_Date' card is showing '15/09/2025' as the selected value. The 'Subject_Code' card is showing 'CS101' as the selected value. The 'Marks_Obtain...' card is showing '35' as the selected value. The 'Max_Marks' card is showing '100' as the selected value. The 'Enrollment_Sta...' card is showing 'Active' as the selected value. The 'Student_ID' card is showing '1301' as the selected value. The 'Student_Name' card is showing 'AARAV SHARMA' as the selected value. The 'Assessment_Type' card is showing 'Final' as the selected value. The 'Exam_Date' card is showing '15/09/2025' as the selected value. The 'Subject_Code' card is showing 'CS101' as the selected value. The 'Marks_Obtain...' card is showing '35' as the selected value. The 'Max_Marks' card is showing '100' as the selected value. The 'Enrollment_Sta...' card is showing 'Active' as the selected value. The 'Student_ID' card is showing '1301' as the selected value. The 'Student_Name' card is showing 'AARAV SHARMA' as the selected value.

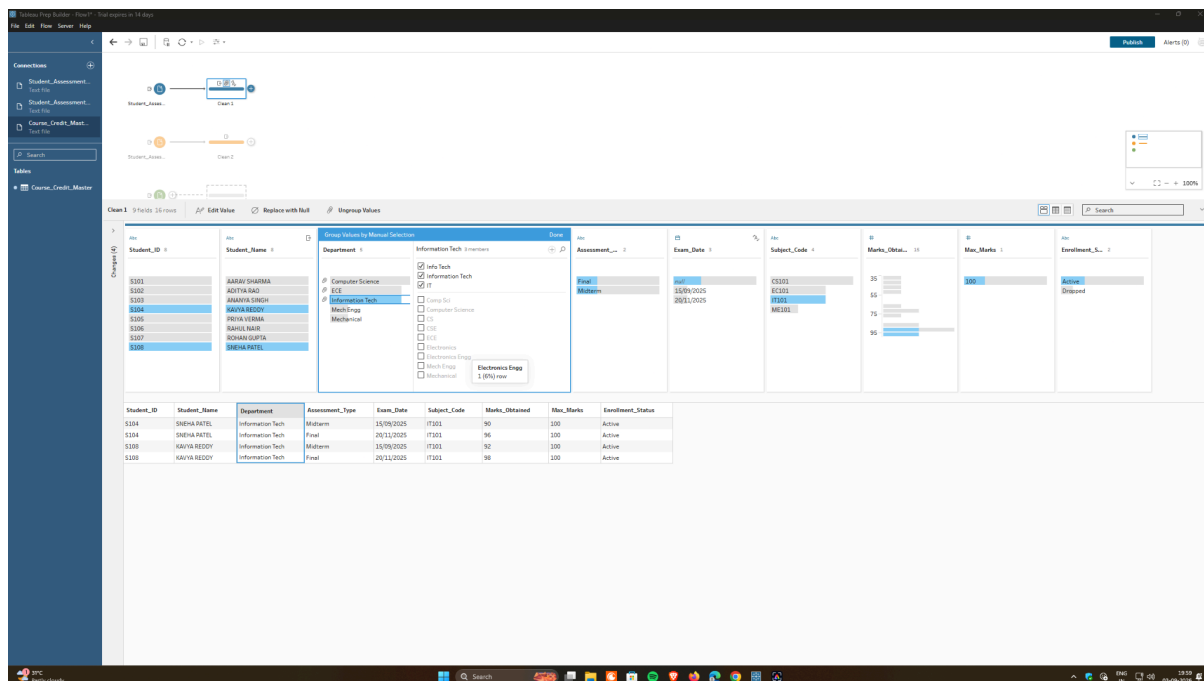
Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtain...	Max_Marks	Enrollment_Sta...
1301	AARAV SHARMA	Computer Science	Midterm	15/09/2025	CS101	35	100	Active
1302	AARAV SHARMA	Computer Science	Final	20/11/2025	CS101	92	100	Active
1303	ROHAN GUPTA	Computer Science	Midterm	15/09/2025	CS101	46	100	Active
1304	ROHAN GUPTA	Computer Science	Final	20/11/2025	CS101	92	100	Active
1305	ANANYA SINGH	Computer Science	Midterm	15/09/2025	CS101	88	100	Active
1306	ANANYA SINGH	Computer Science	Final	20/11/2025	CS101	94	100	Active

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Click on Electronics (or rename the master value to Electronics & Communication). Check the boxes for ECE and Electronics Engg.

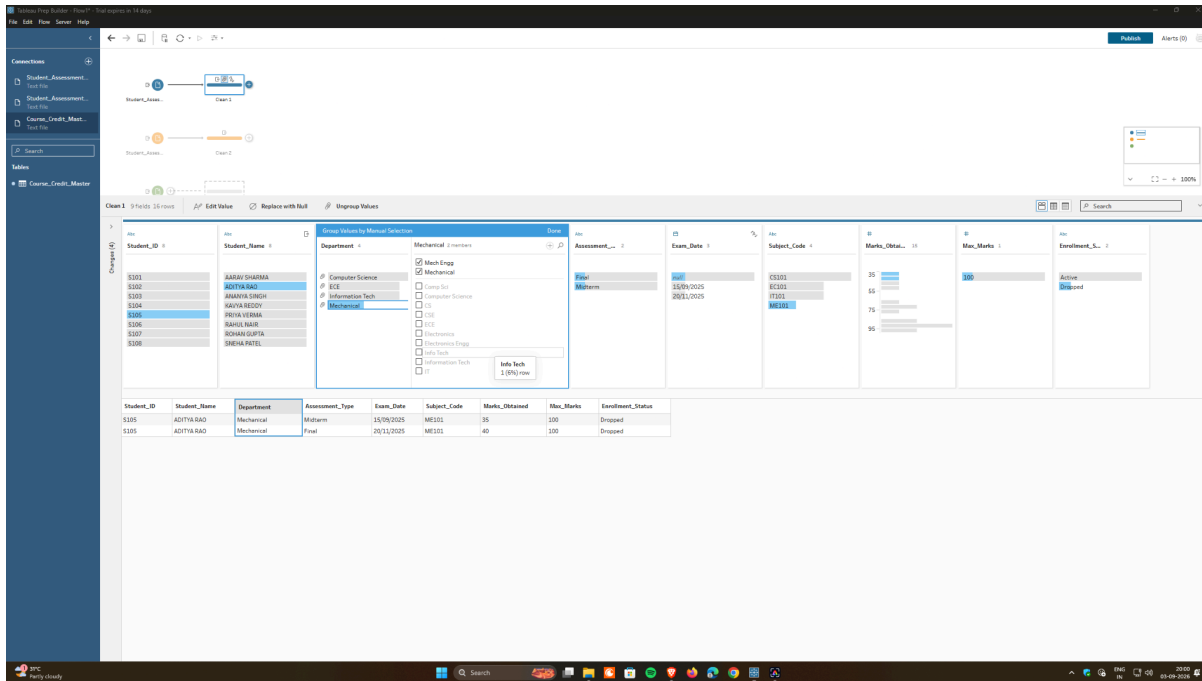


Click on Information Tech (or rename to Information Technology). Check the boxes for IT and Info Tech.

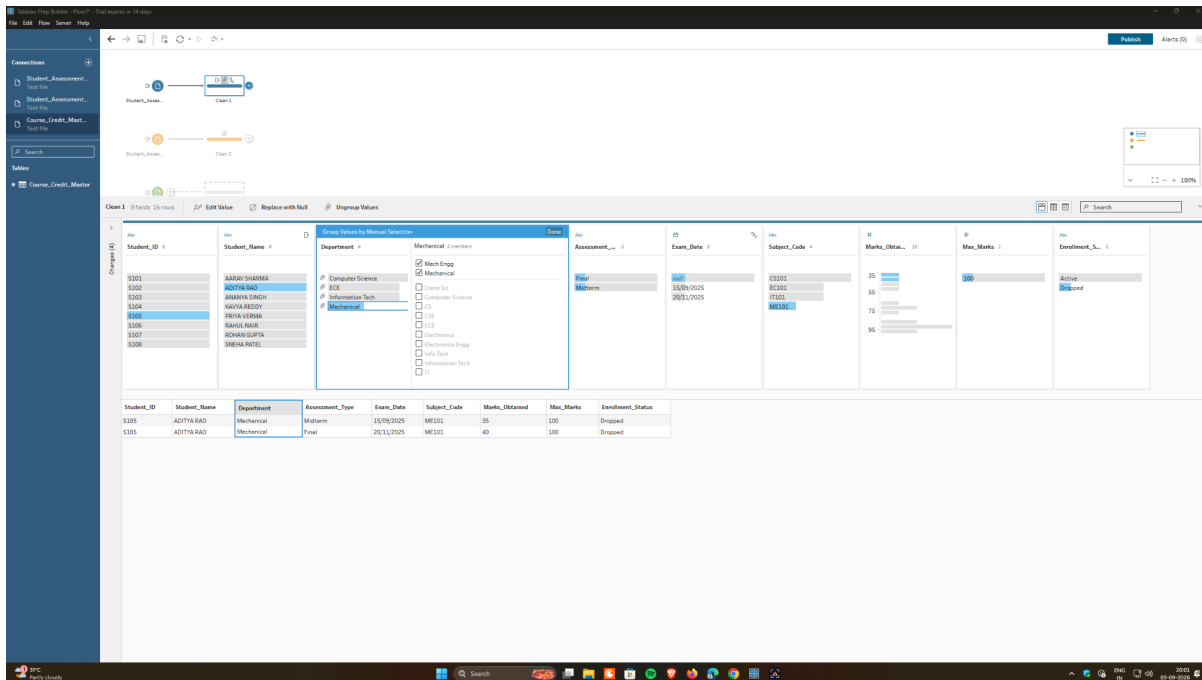


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Click on Mechanical (or rename to Mechanical Engineering). Check the box for Mech Engg.



Click **Done**, then verify the **paperclip icon** appears and the 12 messy department entries are consolidated into **4 clean departments**.



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Step 3: In **Clean 2**, use **More options (...)** → **Group Values** → **Manual Selection** to group **CS, CSE, Computer Science** → **Computer Science**; **ECE, Electronics** → **Electronics**; **IT, Information Tech** → **Information Tech**, then click **Done**.

Standardization Mapping Table:

Original Input Variations	Target Standardized Value	Department Code
Computer Science, Comp Sci, CS, CSE	Computer Science	CS
Electronics, ECE, Electronics Engg	Electronics	EC
Information Tech, IT, Info Tech	Information Tech	IT
Mechanical, Mech Engg	Mechanical	ME

D: Merge Mismatched Fields

Objective: Perform a vertical Union of **Clean 1** (Term 1) and **Clean 2** (Term 2), resolve mismatched column headers (such as **Std_ID** vs **Student_ID**, **Score** vs **Marks_Obtained**), and unify them into a single schema.

When combining datasets originating from different semesters, departments, or source systems, identical information often resides under dissimilar column names. In Tableau Prep, unmerged fields create sparse datasets with duplicate null columns. Merging mismatched fields consolidates those columns without requiring upstream source file modifications.

Step 1: Drag **Clean 2** onto **Clean 1**, select **Union** from the drop zone, and verify that **Union 1** is created connecting both branches.

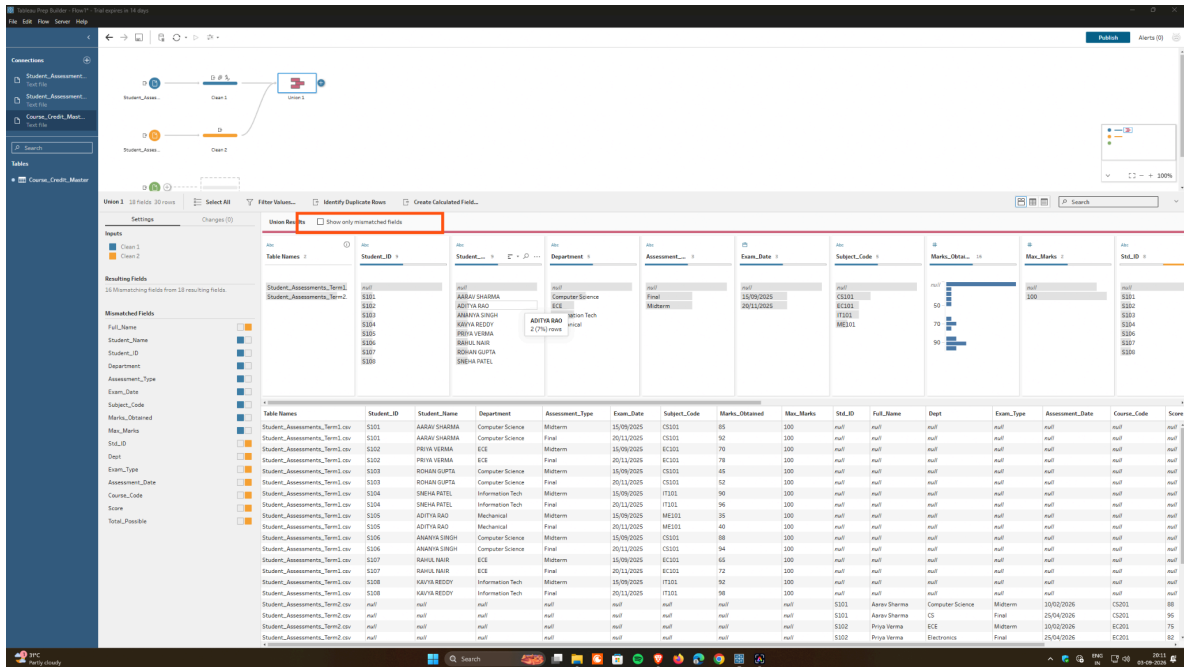
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Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S101	AARAV SHARMA	Computer Science	Final	15/09/2025	CS101	85	100	Active
S102	ADITYA RAO	ECE	Final	15/09/2025	EC101	70	100	Active
S103	ANANYA SINGH	Information Tech	Final	20/11/2025	IT101	78	100	Active
S104	KAVYA REDDY	Computer Science	Final	20/11/2025	CS101	92	100	Active
S105	PRIVA VERMA	Information Tech	Final	15/09/2025	IT101	90	100	Active
S106	RAHUL NAIR	Computer Science	Final	15/09/2025	CS101	96	100	Active
S107	ROHAN GUPTA	Information Tech	Final	15/09/2025	IT101	94	100	Active
S108	SNEHA PATEL	Information Tech	Final	15/09/2025	IT101	98	100	Active
S109	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	35	100	Dropped
S110	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	40	100	Dropped

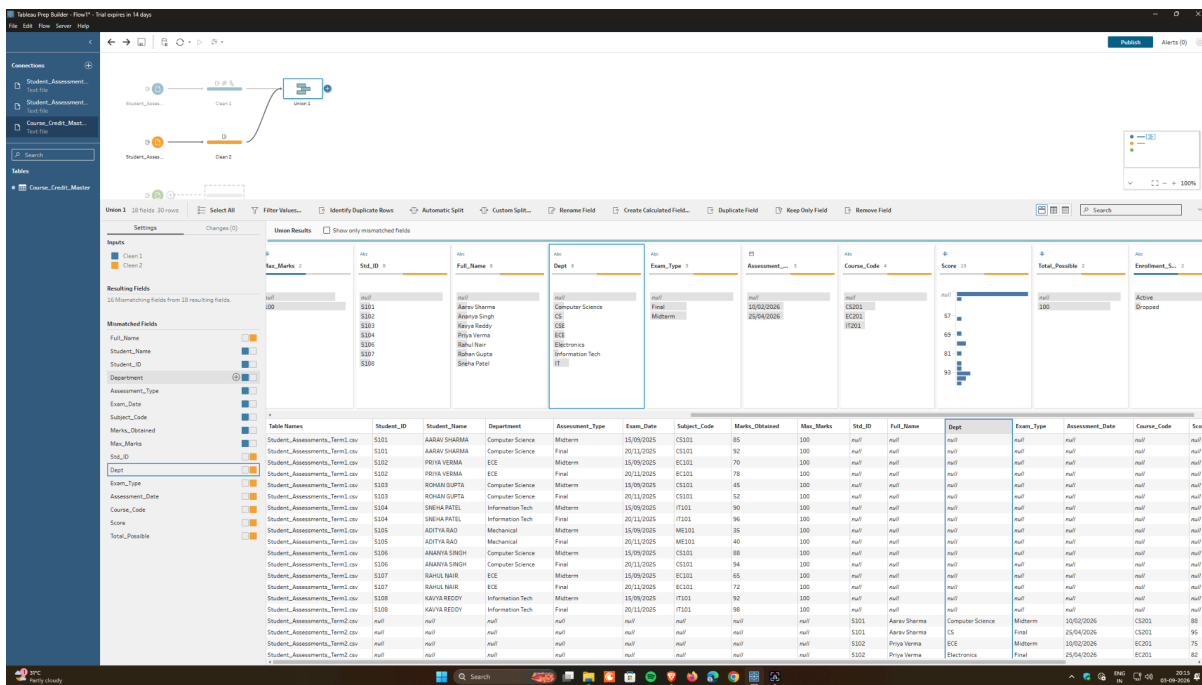
Table Name	Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
Student_Assessments_Term1	S101	AARAV SHARMA	Computer Science	Final	15/09/2025	CS101	85	100	Active
Student_Assessments_Term2	S101	ADITYA RAO	Computer Science	Final	15/09/2025	CS101	70	100	Active
Student_Assessments_Term1	S102	ANANYA SINGH	Information Tech	Final	20/11/2025	IT101	78	100	Active
Student_Assessments_Term2	S102	PRIVA VERMA	ECE	Final	20/11/2025	EC101	92	100	Active
Student_Assessments_Term1	S103	ROHAN GUPTA	Computer Science	Final	15/09/2025	CS101	96	100	Active
Student_Assessments_Term2	S103	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	35	100	Dropped
Student_Assessments_Term1	S104	SNEHA PATEL	Information Tech	Final	15/09/2025	IT101	98	100	Active
Student_Assessments_Term2	S104	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	40	100	Dropped
Student_Assessments_Term1	S105	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	35	100	Dropped
Student_Assessments_Term2	S105	PRIVA VERMA	ECE	Final	20/11/2025	EC101	92	100	Active
Student_Assessments_Term1	S106	ANANYA SINGH	Information Tech	Final	15/09/2025	IT101	90	100	Active
Student_Assessments_Term2	S106	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	35	100	Dropped
Student_Assessments_Term1	S107	RAHUL NAIR	Computer Science	Final	15/09/2025	CS101	96	100	Active
Student_Assessments_Term2	S107	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	40	100	Dropped
Student_Assessments_Term1	S108	KAVYA REDDY	Information Tech	Final	15/09/2025	IT101	94	100	Active
Student_Assessments_Term2	S108	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	35	100	Dropped
Student_Assessments_Term1	S109	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	35	100	Dropped
Student_Assessments_Term2	S109	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	40	100	Dropped
Student_Assessments_Term1	S110	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	40	100	Dropped
Student_Assessments_Term2	S110	ADITYA RAO	Mechanical	Final	20/11/2025	ME101	40	100	Dropped

Step 2: Select **Union 1** and review the **Mismatched Fields** section to identify differently named but equivalent fields, such as **Student_ID/Std_ID**, **Student_Name/Full_Name**, **Department/Dept**, **Assessment_Type/Exam_Type**, **Exam_Date/Assessment_Date**, **Subject_Code/Course_Code**, **Marks_Obtained/Score**, and **Max_Marks/Total_Possible**.

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Step 3: Drag each mismatched field onto its corresponding field to merge **Std_ID**→**Student_ID**, **Full_Name**→**Student_Name**, **Dept**→**Department**, **Assessment_Date**→**Exam_Date**, **Course_Code**→**Subject_Code**, **Score**→**Marks_Obtained**, **Total_Possible**→**Max_Marks**, and **Exam_Type**→**Assessment_Type**, until the mismatch count becomes 0.



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Ass_Marks	Std_ID	Full_Name	Dept	Exam_Type	Assessment...	Course_Code	Score	Total_Possible	Enrollment_S...
88	S001	AARAV SHARMA	Computer Science	Final	15/09/2025	CS001	85	100	Active
92	S001	AARAV SHARMA	Computer Science	Final	20/11/2025	CS001	92	100	Active
70	S002	PRIVIA VERMA	ECE	Final	15/09/2025	EC001	70	100	Active
78	S002	PRIVIA VERMA	ECE	Final	20/11/2025	EC001	78	100	Active
45	S003	ROMAN GUPTA	Computer Science	Final	15/09/2025	CS001	45	100	Active
52	S003	ROMAN GUPTA	Computer Science	Final	20/11/2025	CS001	52	100	Active
90	S004	SNEHA RATEL	Information Tech	Final	15/09/2025	IT001	90	100	Active
96	S004	SNEHA RATEL	Information Tech	Final	20/11/2025	IT001	96	100	Active
36	S005	ADITYA RAO	Mechanical	Final	15/09/2025	ME001	36	100	Active
40	S005	ADITYA RAO	Mechanical	Final	20/11/2025	ME001	40	100	Active
88	S006	ANANYA SINGH	Computer Science	Final	15/09/2025	CS001	88	100	Active
94	S006	ANANYA SINGH	Computer Science	Final	20/11/2025	CS001	94	100	Active
65	S007	RAHUL NAIR	ECE	Final	15/09/2025	EC001	65	100	Active
72	S007	RAHUL NAIR	ECE	Final	20/11/2025	EC001	72	100	Active
94	S008	KAVYA REDDY	Information Tech	Final	15/09/2025	IT001	94	100	Active
96	S008	KAVYA REDDY	Information Tech	Final	20/11/2025	IT001	96	100	Active
88	S001	AARAV SHARMA	Computer Science	Final	15/09/2025	CS001	88	100	Active
96	S001	AARAV SHARMA	Computer Science	Final	20/11/2025	CS001	96	100	Active
75	S002	PRIVIA VERMA	ECE	Final	15/09/2025	EC001	75	100	Active
82	S002	PRIVIA VERMA	ECE	Final	20/11/2025	EC001	82	100	Active

Student_Name	Student_ID	Department	Std_ID	Dept	Enrollment_Status	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_S...
AARAV SHARMA	S001	Computer Science	CS001	Computer Science	Active	Final	15/09/2025	CS001	85	100	Active
AARAV SHARMA	S001	Computer Science	CS001	Computer Science	Active	Final	20/11/2025	CS001	92	100	Active
PRIVIA VERMA	S002	ECE	EC001	ECE	Active	Final	15/09/2025	EC001	70	100	Active
PRIVIA VERMA	S002	ECE	EC001	ECE	Active	Final	20/11/2025	EC001	78	100	Active
ROMAN GUPTA	S003	Computer Science	CS001	Computer Science	Active	Final	15/09/2025	CS001	45	100	Active
ROMAN GUPTA	S003	Computer Science	CS001	Computer Science	Active	Final	20/11/2025	CS001	52	100	Active
SNEHA RATEL	S004	Information Tech	IT001	Information Tech	Active	Final	15/09/2025	IT001	90	100	Active
SNEHA RATEL	S004	Information Tech	IT001	Information Tech	Active	Final	20/11/2025	IT001	96	100	Active
ADITYA RAO	S005	Mechanical	ME001	Mechanical	Dropped	Final	15/09/2025	ME001	36	100	Active
ADITYA RAO	S005	Mechanical	ME001	Mechanical	Dropped	Final	20/11/2025	ME001	40	100	Active
ANANYA SINGH	S006	Computer Science	CS001	Computer Science	Active	Final	15/09/2025	CS001	88	100	Active
ANANYA SINGH	S006	Computer Science	CS001	Computer Science	Active	Final	20/11/2025	CS001	94	100	Active
RAHUL NAIR	S007	ECE	EC001	ECE	Active	Final	15/09/2025	EC001	65	100	Active
RAHUL NAIR	S007	ECE	EC001	ECE	Active	Final	20/11/2025	EC001	72	100	Active
KAVYA REDDY	S008	Information Tech	IT001	Information Tech	Active	Final	15/09/2025	IT001	94	100	Active
KAVYA REDDY	S008	Information Tech	IT001	Information Tech	Active	Final	20/11/2025	IT001	96	100	Active
AARAV SHARMA	S001	Computer Science	CS001	Computer Science	Active	Final	15/09/2025	CS001	88	100	Active
AARAV SHARMA	S001	Computer Science	CS001	Computer Science	Active	Final	20/11/2025	CS001	96	100	Active
PRIVIA VERMA	S002	ECE	EC001	ECE	Active	Final	15/09/2025	EC001	75	100	Active
PRIVIA VERMA	S002	ECE	EC001	ECE	Active	Final	20/11/2025	EC001	82	100	Active

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Canvas View: Clean 1, Clean 2, Union 1

Settings: Clean 1, Clean 2

Resulting Fields: 10 resulting fields from 10 resulting fields.

Table Names	Enrollment_Status	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Dept	Std_ID
Pharma	Active	Final	15/09/2025	CS101	85	100	Computer Science	S101
SHARMA	Active	Final	20/11/2025	CS101	90	100	Computer Science	S102
I.RAO	Active	Final	15/09/2025	EC101	70	100	ECE	S103
V.SINGH	Active	Final	20/11/2025	EC101	70	100	ECE	S102
V.SINGH	Active	Final	15/09/2025	CS101	45	100	Computer Science	S103
REDIY	Active	Final	20/11/2025	CS101	52	100	Computer Science	S103
KIRMA	Active	Final	15/09/2025	IT101	90	100	Information Tech	S104
JERMA	Active	Final	20/11/2025	IT101	90	100	Information Tech	S104
KIR	Dropped	Midterm	15/09/2025	ME101	35	100	Mechanical	S105
NAIR	Dropped	Final	20/11/2025	ME101	40	100	Mechanical	S105
Gupta	Active	Final	15/09/2025	CS101	88	100	Computer Science	S106
	Active	Final	20/11/2025	CS101	94	100	Computer Science	S106
	Active	Final	15/09/2025	EC101	45	100	ECE	S107
	Active	Final	20/11/2025	EC101	72	100	ECE	S107
	Active	Final	15/09/2025	IT101	90	100	Information Tech	S108
	Active	Final	20/11/2025	IT101	90	100	Information Tech	S108
	Active	Final	15/09/2025	CS101	88	100	Computer Science	S101
	Active	Final	20/11/2025	CS101	95	100	CS	S101
	Active	Final	15/09/2025	EC101	75	100	ECE	S102
	Active	Final	20/11/2025	EC101	82	100	Electronics	S102

Step 4: Click + next to **Union 1**, select **Clean Step** (creating **Clean 3**; it may appear as **Clean 4** if another Clean Step already exists), and verify **30 total rows** across **10 unified columns**, including **Table Names**

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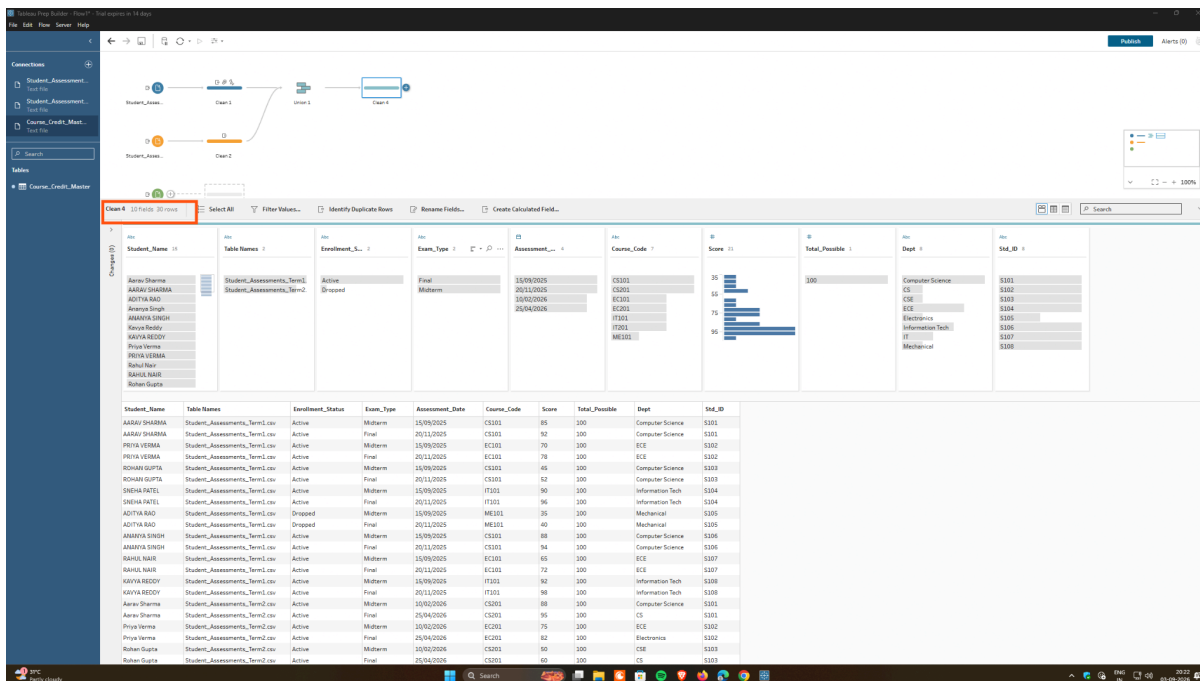
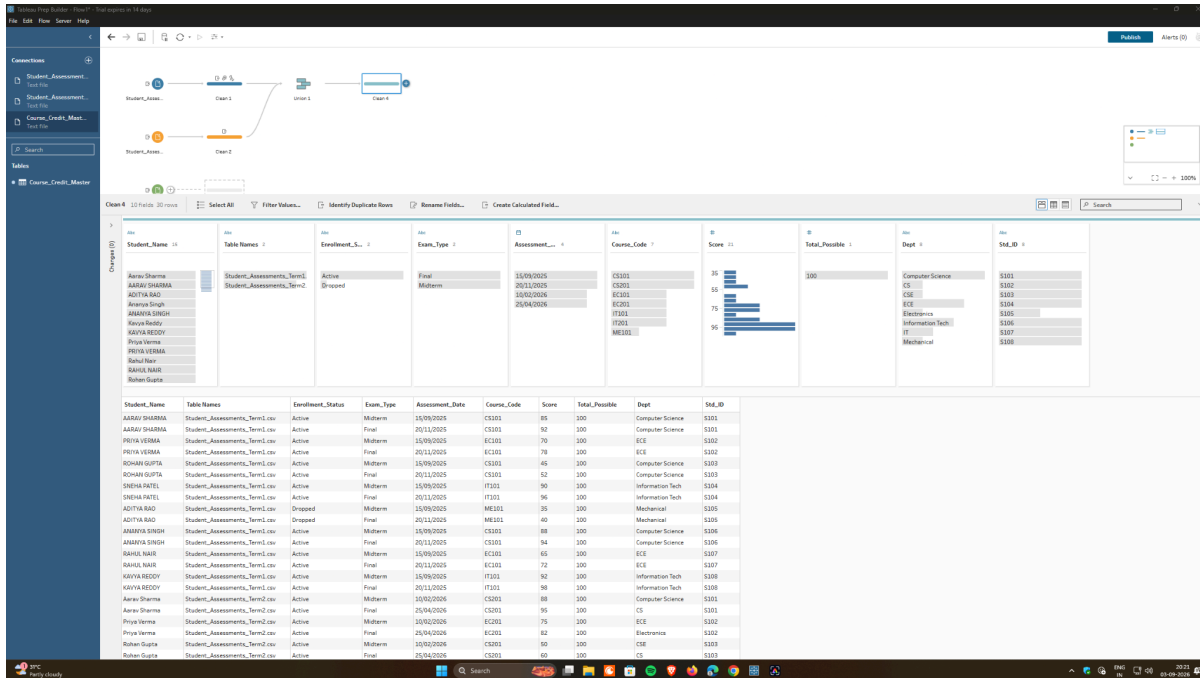
Canvas View: Clean 1, Clean 2, Union 1, Clean 3

Settings: Clean 1, Clean 2

Resulting Fields: 10 resulting fields from 10 resulting fields.

Table Names	Enrollment_Status	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Dept	Std_ID
Pharma	Active	Final	15/09/2025	CS101	85	100	Computer Science	S101
SHARMA	Active	Final	20/11/2025	CS101	90	100	Computer Science	S102
I.RAO	Active	Final	15/09/2025	EC101	70	100	ECE	S103
V.SINGH	Active	Final	20/11/2025	EC101	70	100	ECE	S102
V.SINGH	Active	Final	15/09/2025	CS101	45	100	Computer Science	S103
REDIY	Active	Final	20/11/2025	CS101	52	100	Computer Science	S103
KIRMA	Active	Final	15/09/2025	IT101	90	100	Information Tech	S104
JERMA	Active	Final	20/11/2025	IT101	90	100	Information Tech	S104
KIR	Dropped	Midterm	15/09/2025	ME101	35	100	Mechanical	S105
NAIR	Dropped	Final	20/11/2025	ME101	40	100	Mechanical	S105
Gupta	Active	Final	15/09/2025	CS101	88	100	Computer Science	S106
	Active	Final	20/11/2025	CS101	94	100	Computer Science	S106
	Active	Final	15/09/2025	EC101	45	100	ECE	S107
	Active	Final	20/11/2025	EC101	72	100	ECE	S107
	Active	Final	15/09/2025	IT101	90	100	Information Tech	S108
	Active	Final	20/11/2025	IT101	90	100	Information Tech	S108
	Active	Final	15/09/2025	CS101	88	100	Computer Science	S101
	Active	Final	20/11/2025	CS101	95	100	CS	S101
	Active	Final	15/09/2025	EC101	75	100	ECE	S102
	Active	Final	20/11/2025	EC101	82	100	Electronics	S102

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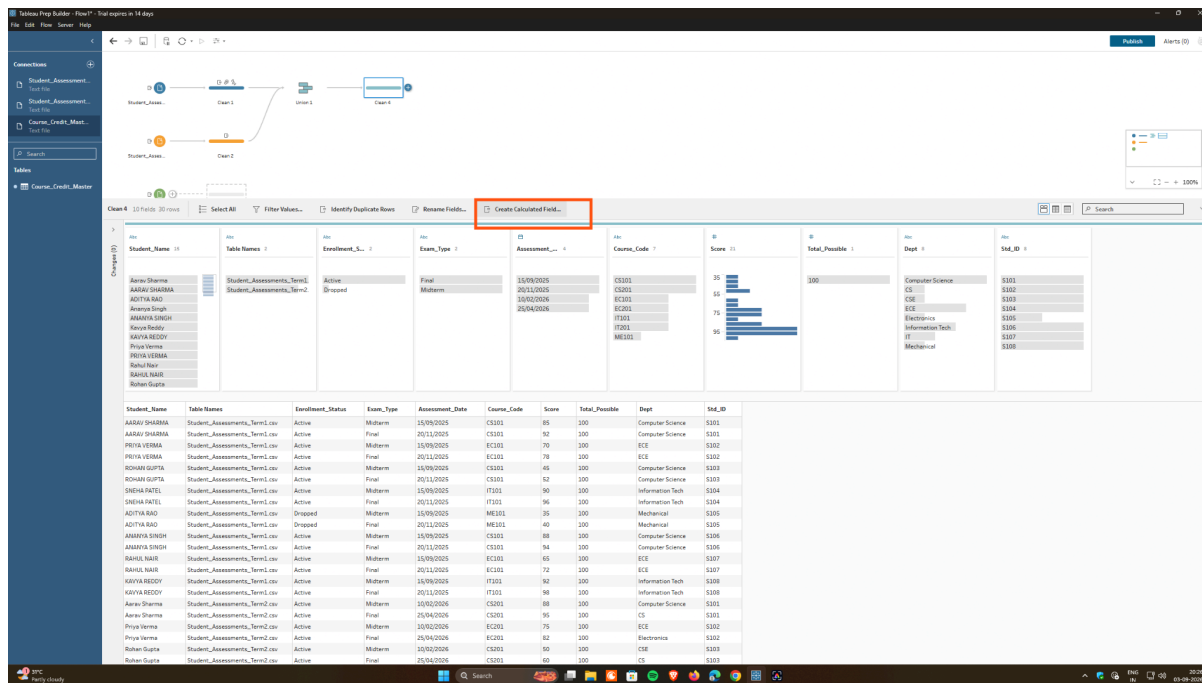
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E: Perform Aggregations in Prep

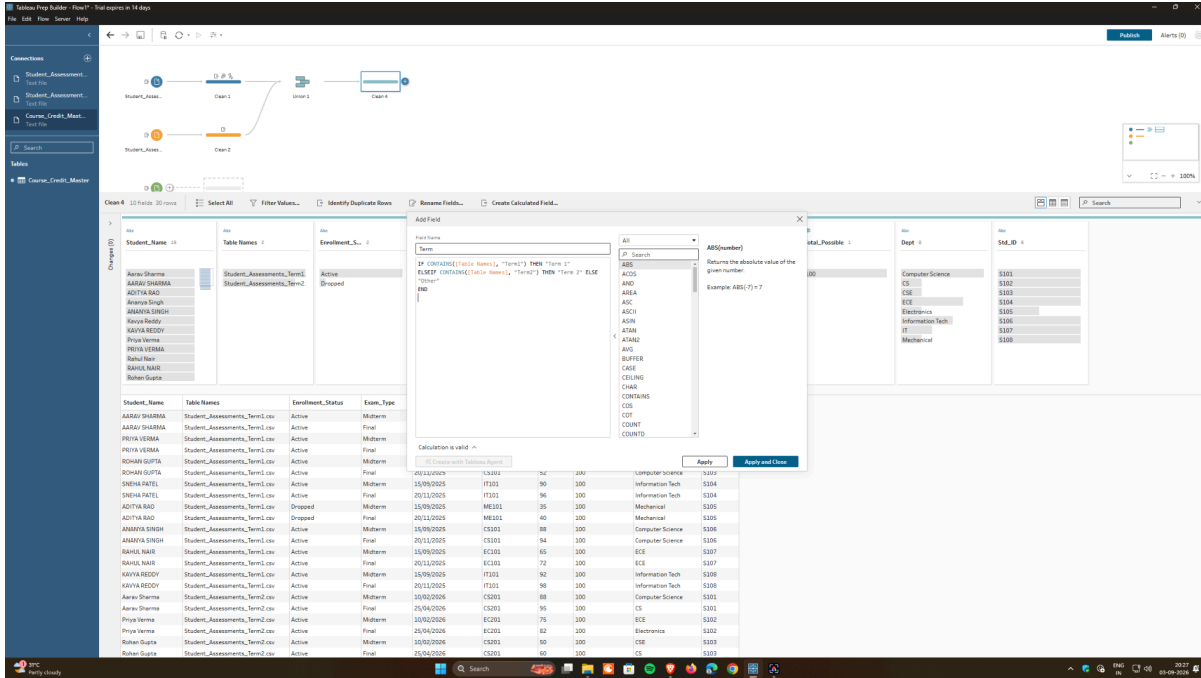
Objective: Create an Aggregate Step in Tableau Prep to summarize granular, exam-level test scores into student-level cumulative metrics (Total Marks, Average Score, and Total Assessment Count) per term

Raw assessment tables log every individual examination (Midterm, Final) on a separate row. By performing an aggregation step directly in Tableau Prep, we roll up individual row events into pre-computed summary statistics, reducing data redundancy and generating student scorecard baselines prior to dashboarding.

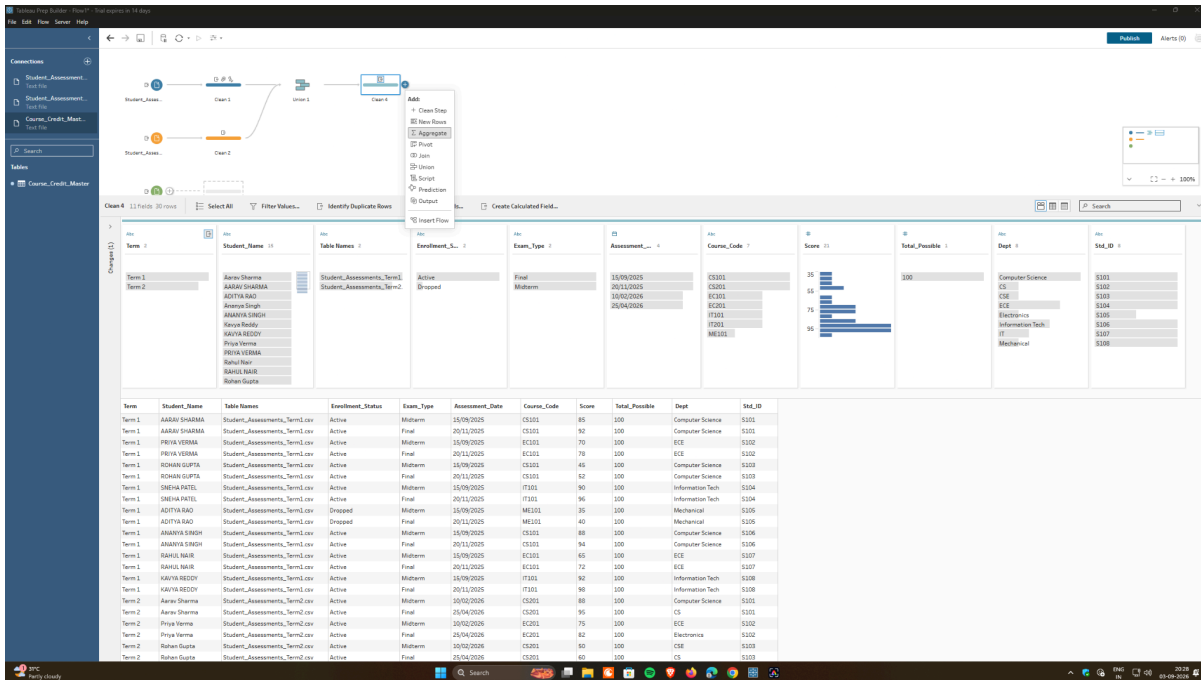
Step 1: In Clean 3, use More options (...) → Create Calculated Field, name it Term, enter the given **CONTAINS([Table Names], "Term1/Term2")** formula to classify records as Term 1, Term 2, or Other, then click Save.



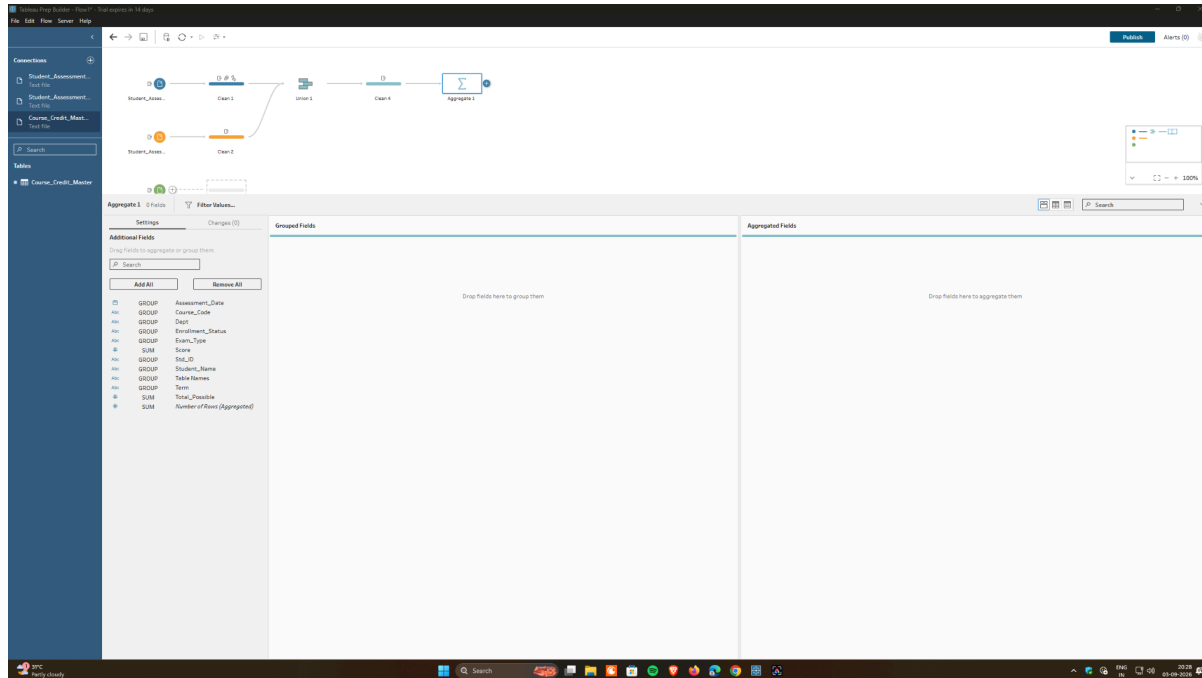
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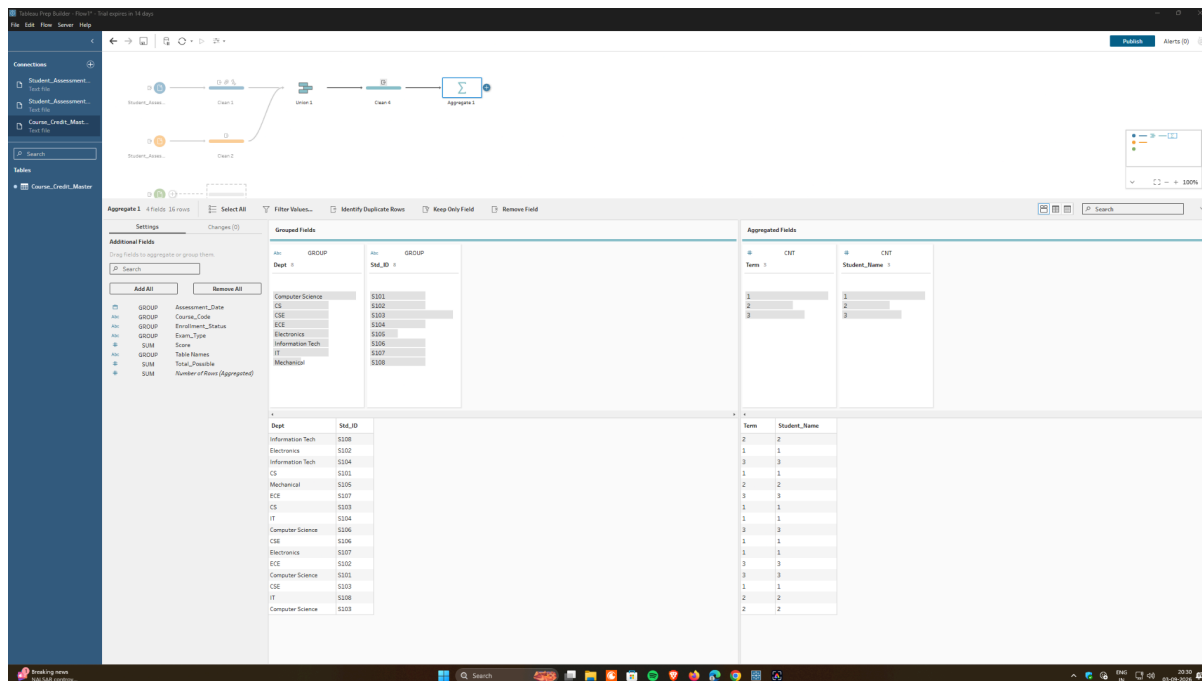
Step 2: Hover over Clean 3, click +, select Aggregate (Σ), and verify that Aggregate 1 appears with Grouped Fields and Aggregated Fields drop zones.



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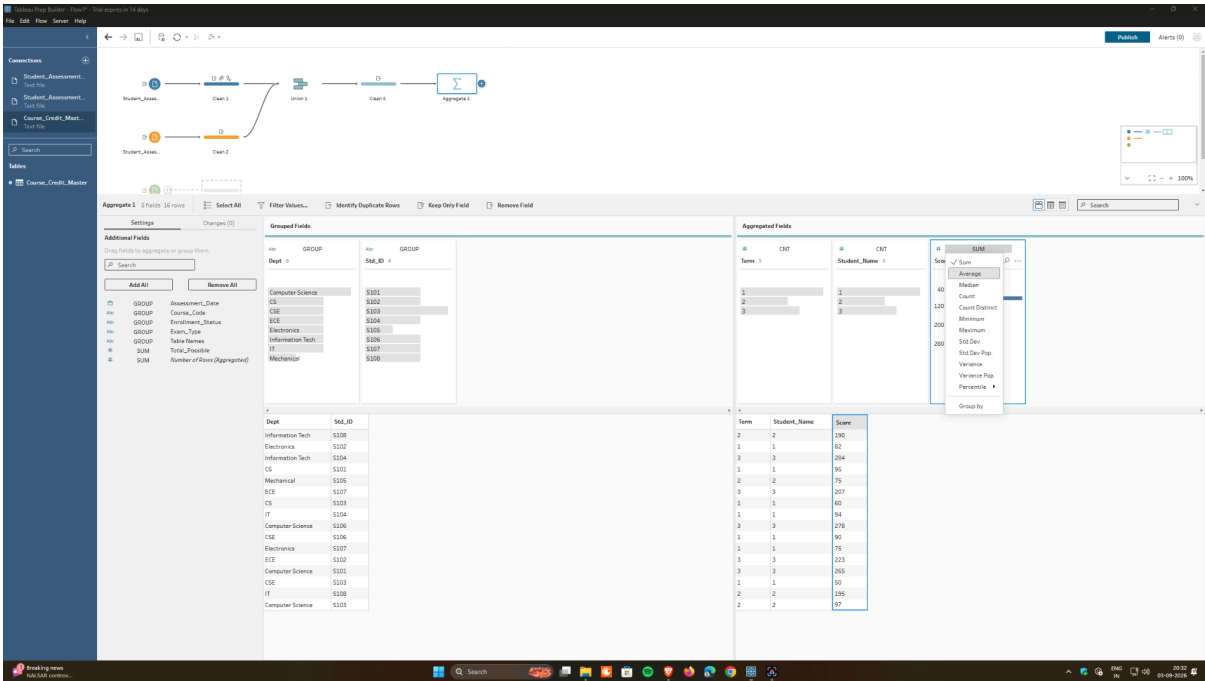
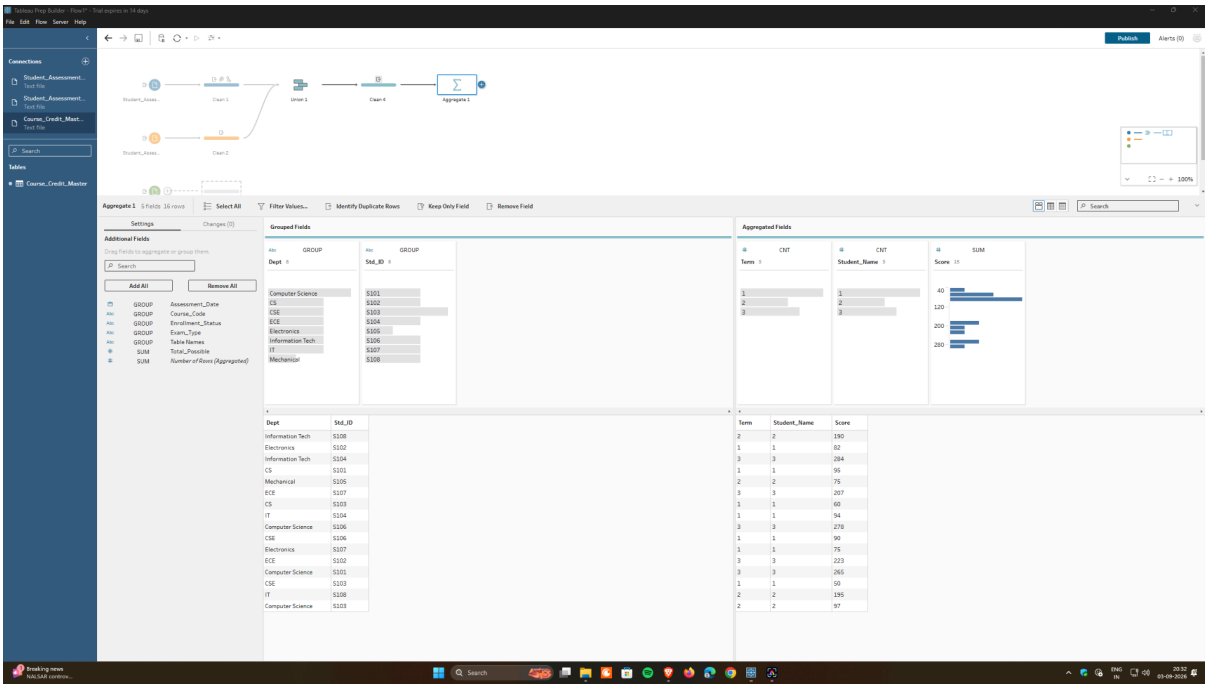


Step 3: In the Aggregate step, drag Student_ID, Student_Name, Department, and Term into the Grouped Fields box.

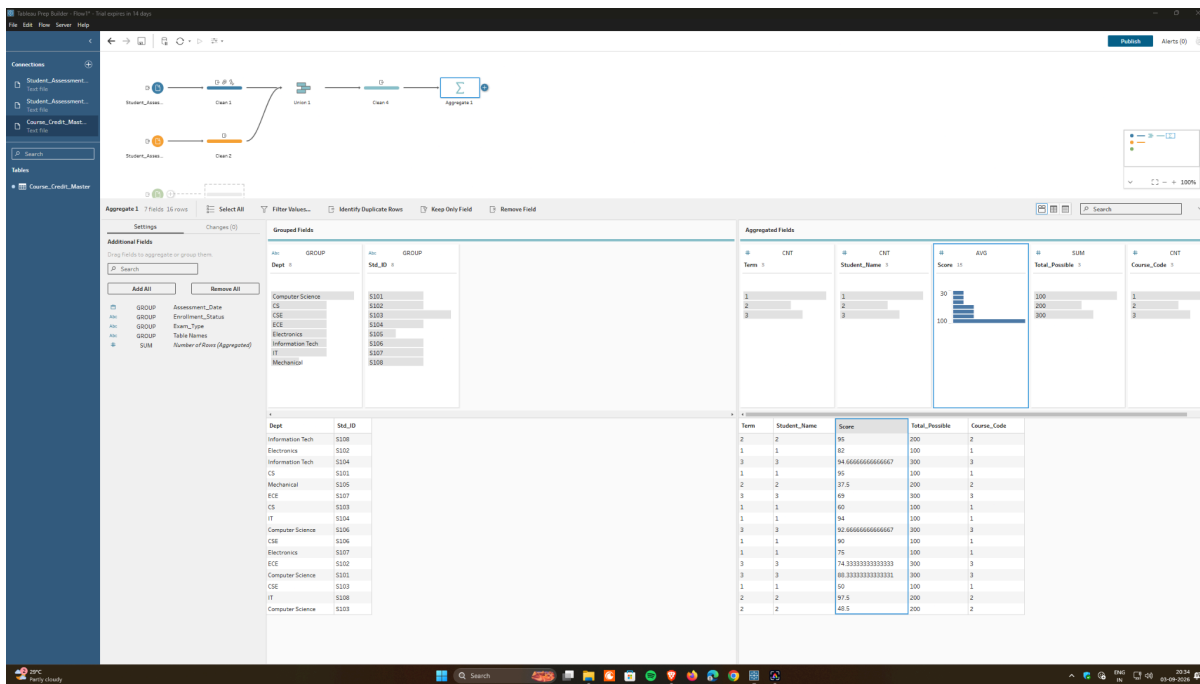
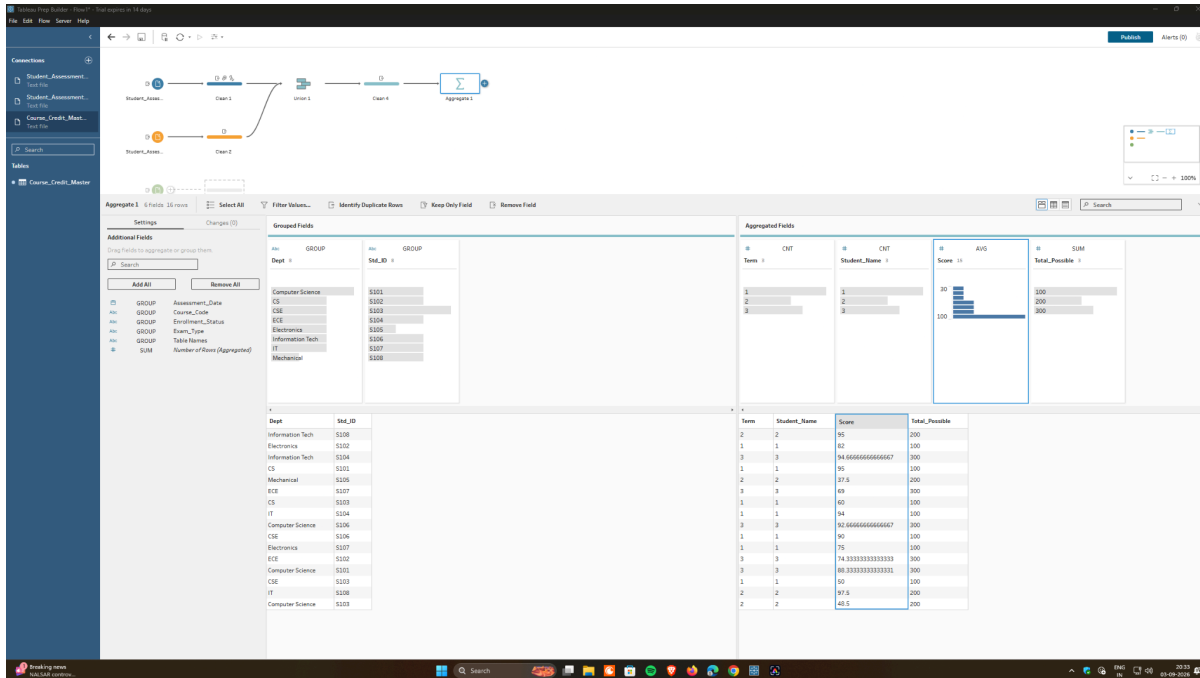


Step 4: In Aggregated Fields, add Marks_Obtained → Sum for Total Marks, use Marks_Obtained → Average for Average Score, and add Max_Marks → Sum for Total Possible Marks.

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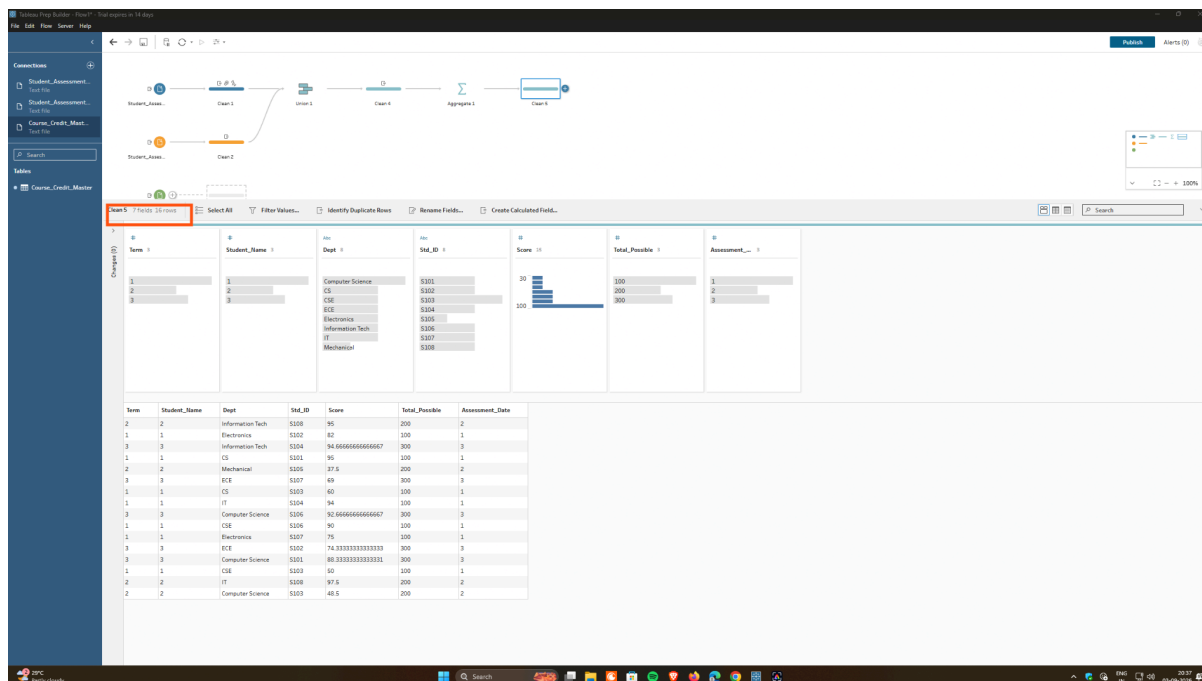
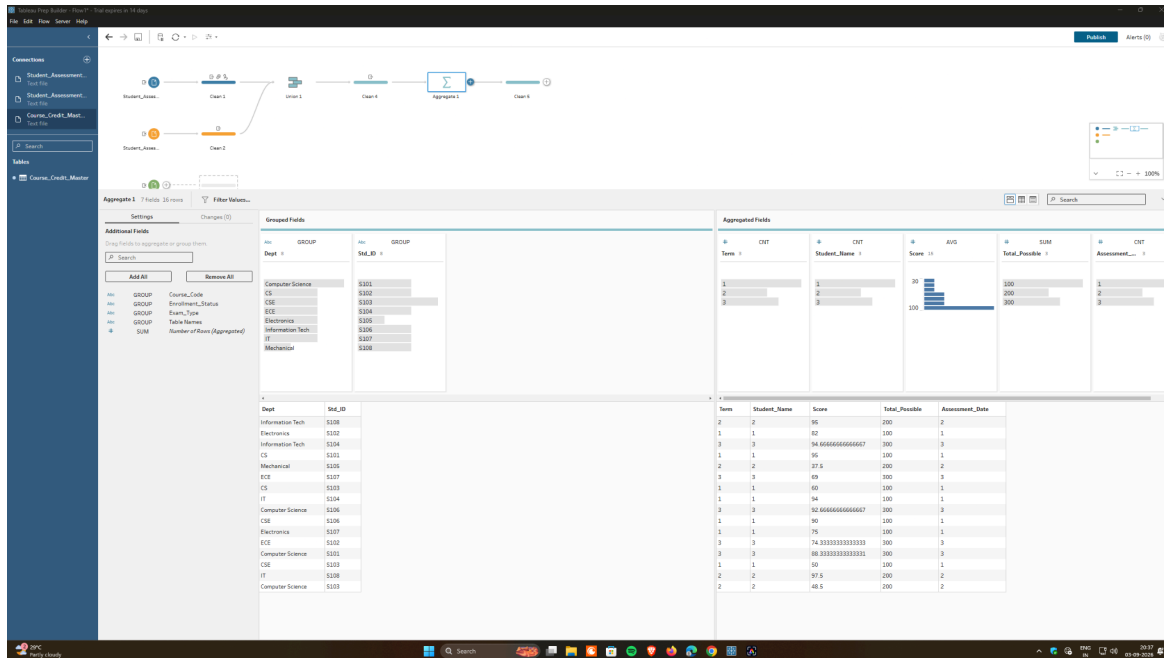
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Step 5: Rename the aggregated fields as follows: **Marks_Obtained (Sum)** → **Total_Marks_Obtained**, **Marks_Obtained (Average)** → **Average_Score**, **Max_Marks (Sum)** → **Total_Max_Marks**, and **Assessment_Type (Count)** → **Assessments_Taken**.

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Step 6: Click + next to **Aggregate 1**, select **Clean Step** to create **Clean 4**, and verify that the **30 assessment rows** are reduced to about **15 summary records** (one row per student per term).



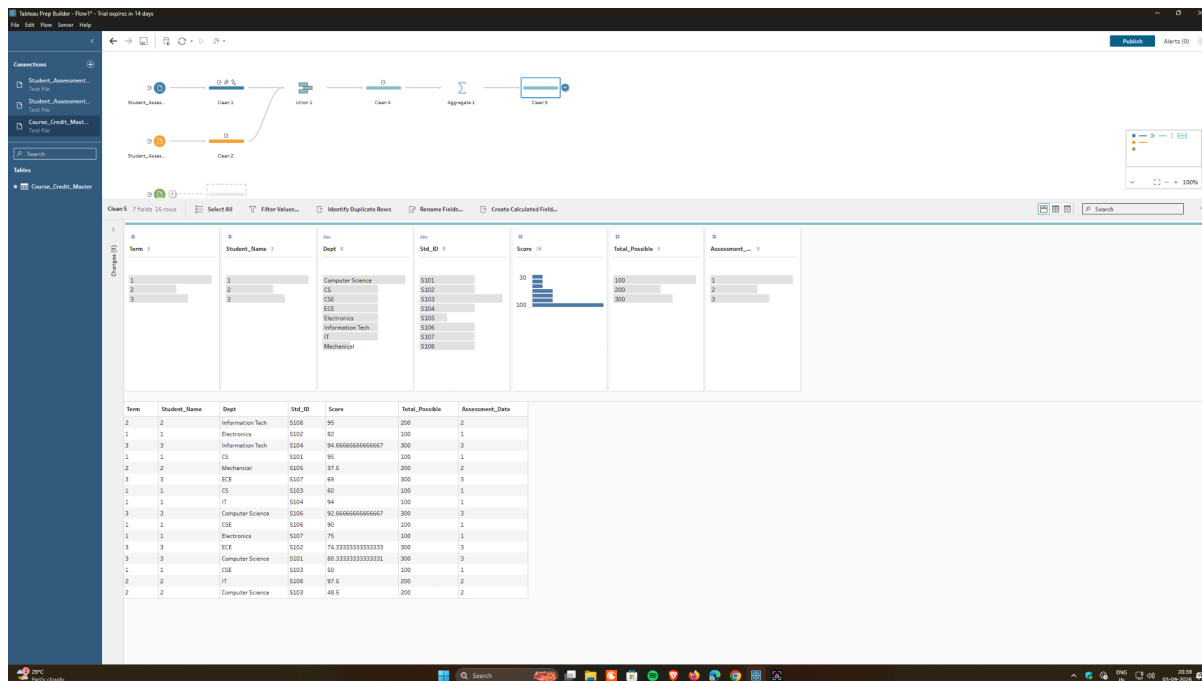
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F: Apply Filters and Transformations

Objective: Remove inactive/dropped records, compute weighted percentages, assign automated grading tiers, and derive academic standing indicators using calculated fields in Tableau Prep.

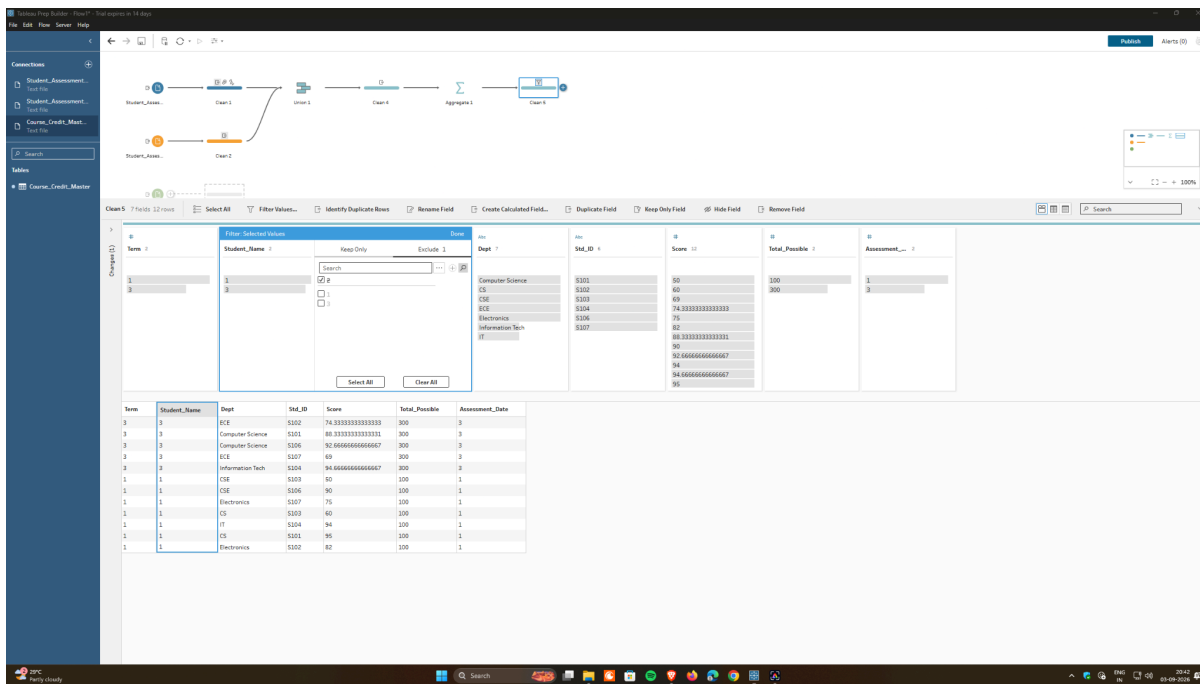
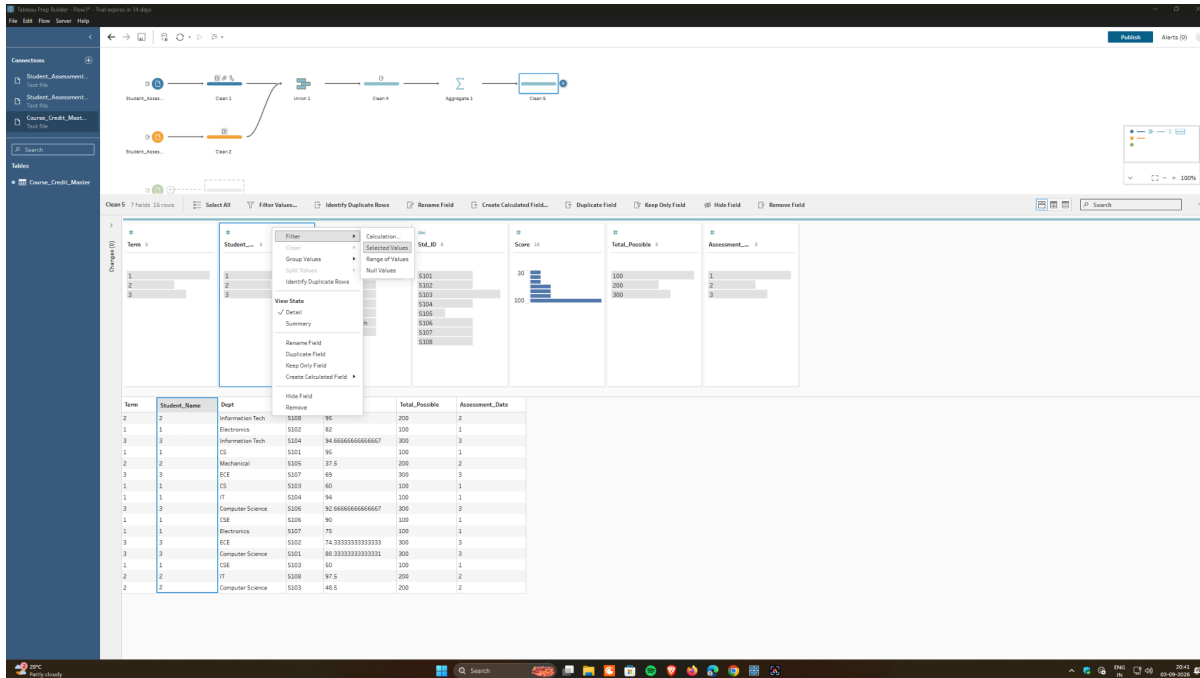
Not all raw records should enter downstream analytical dashboards (e.g., students who dropped out mid-term skew department averages). Transforming raw totals into percentage metrics and standardized letter grades directly inside the data prep pipeline ensures that business logic is applied once and remains consistent across all downstream reporting tools.

Step 1: Select **Clean 4** after **Aggregate 1** and verify the Profile Pane shows the aggregated fields: **Student_ID**, **Student_Name**, **Department**, **Term**, **Total_Marks_Obtained**, **Average_Score**, **Total_Max_Marks**, and **Assessments_Taken** (names may vary based on your Union).



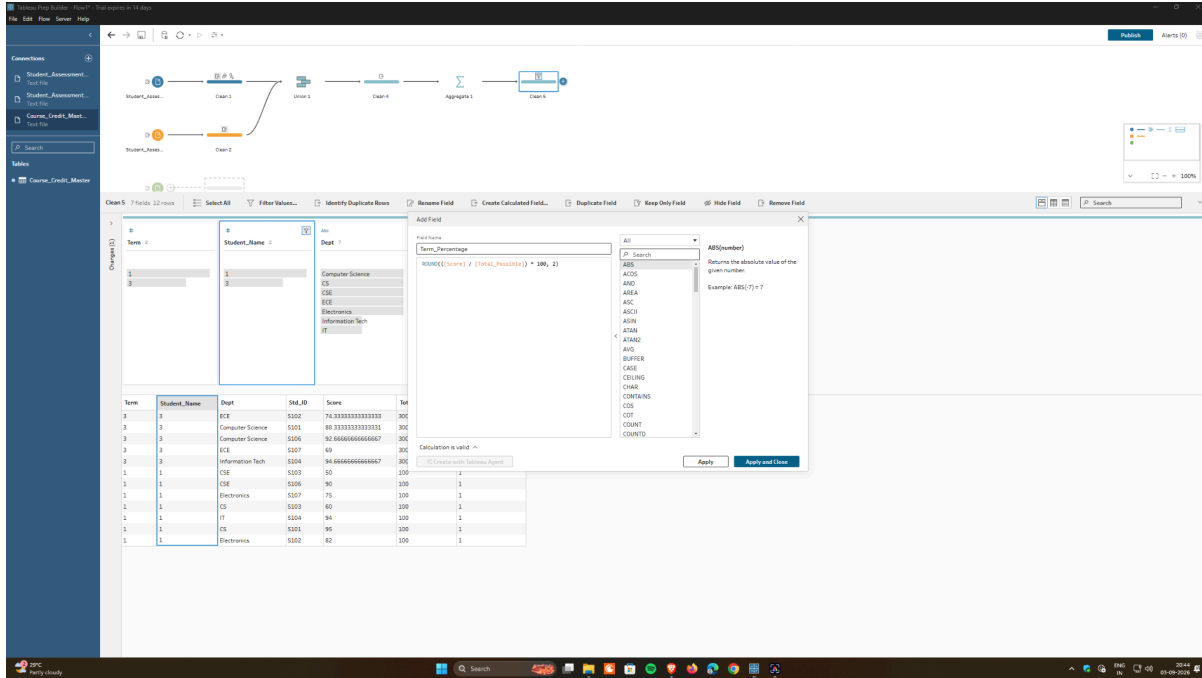
Step 2: If **Enrollment_Status** is available, filter it to keep only **Active** students by excluding **Dropped** records using **Exclude** or **Filter Values** → **Selected Values**.

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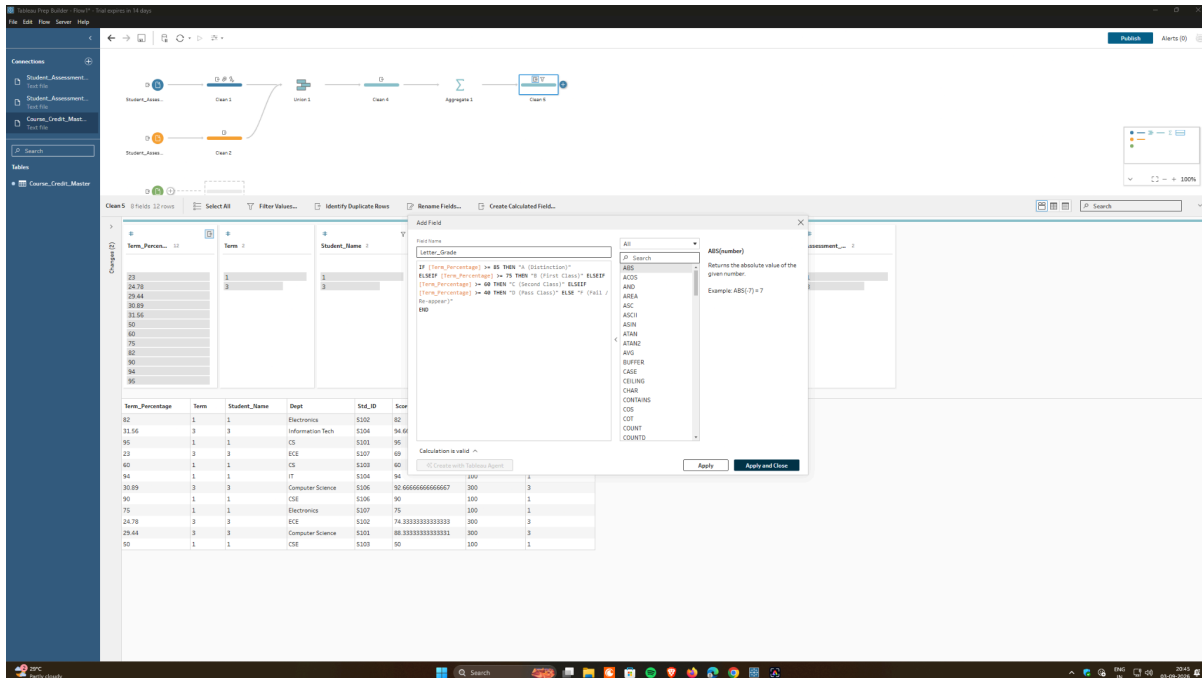


Step 3: If Enrollment_Status is available, filter it to keep only Active students by excluding Dropped records using Exclude or Filter Values → Selected Values.

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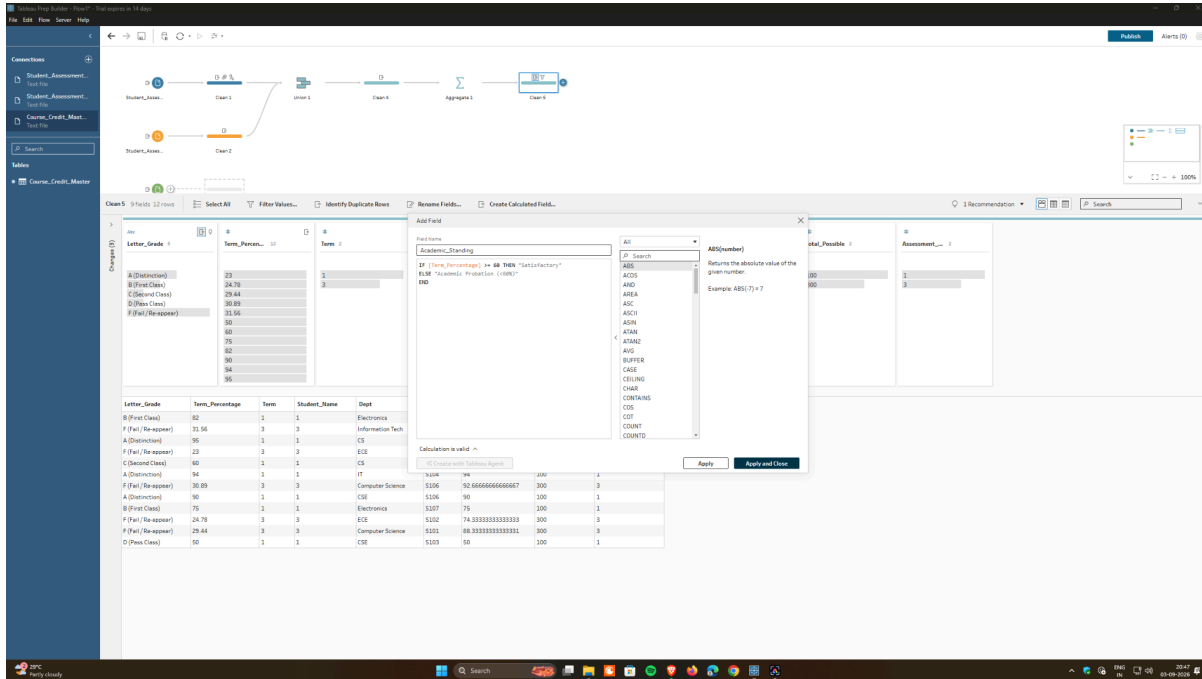


Step 4: Create a calculated field named **Letter_Grade** using the given **Term_Percentage** thresholds to classify students from **A (Distinction)** to **F (Fail / Re-appear)**, then click **Save**.

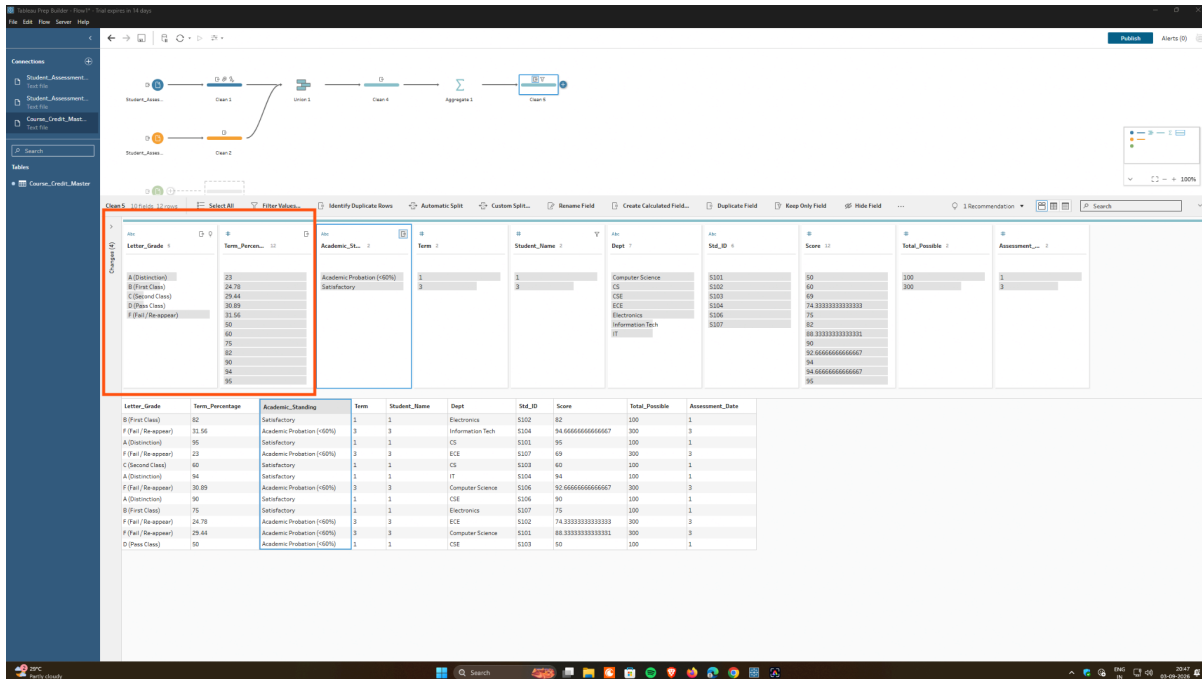


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```
Step 5: Create a calculated field named Academic_Standing using IF
[Term_Percentage] >= 60 THEN "Satisfactory" ELSE "Academic Probation
(<60%)" END.
```



Step 6: Verify Clean 4 to ensure Term_Percentage ranges from 0-100%, Letter_Grade shows categorical grade buckets, and Academic_Standing identifies students needing academic counseling.



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Transformation Summary Table

Transformation	Type	Formula / Rule	Output Meaning
Filter Status	Row Exclusion	[Enrollment_Status] != 'Dropped'	Removes dropped students
Term_Percentage	Numeric Calculation	([Total_Marks_Obtained] / [Total_Max_Marks]) * 100	Normalized 0-100 scale
Letter_Grade	Conditional Dimension	IF [Term_Percentage] >= 85 THEN "A (Distinction)" ...	Standardized letter grade
Academic_Standi ng	Conditional Flag	IF [Term_Percentage] >= 60 THEN "Satisfactory" ...	Compliance warning tag

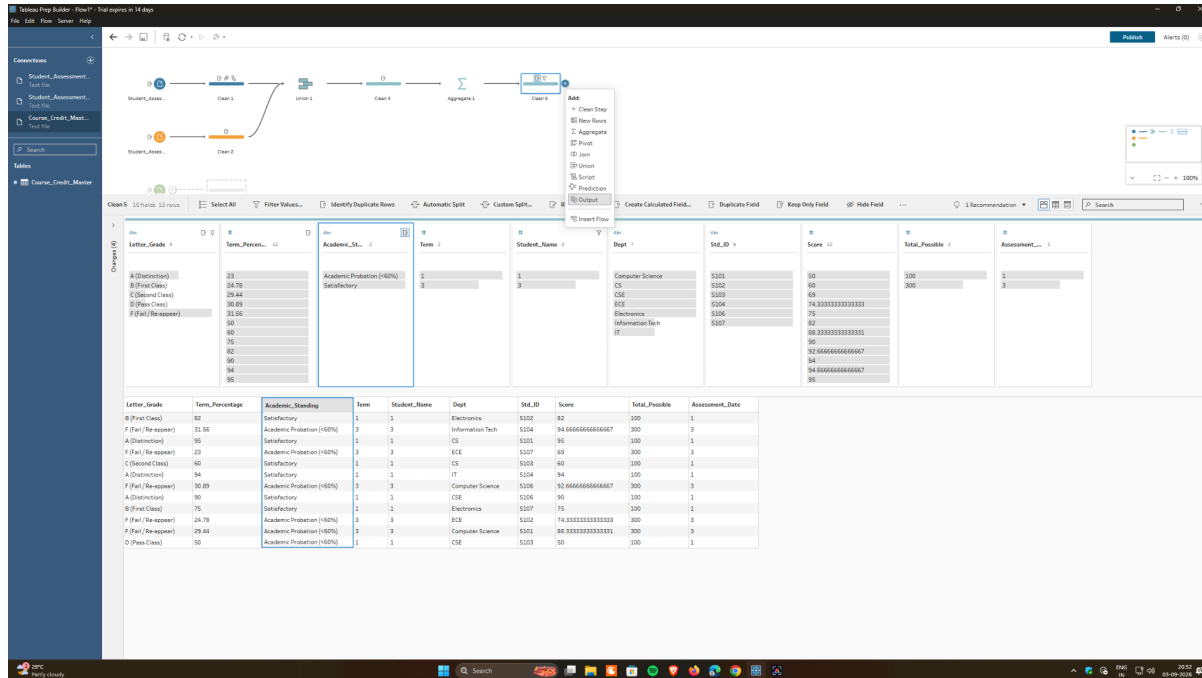
G: Create and Automate Data Preparation Flows

Objective: Configure an Output Step in Tableau Prep Builder to publish the fully transformed pipeline to an extract file (.hyper or .csv), and learn how to automate or execute the flow on demand.

Data preparation workflows must be repeatable and production-ready. Configuring a dedicated output node ensures that whenever new assessment scores arrive, running the flow automatically executes all trimming, grouping, merging, filtering, and scoring calculations in seconds without any manual rework.

Step 1: Hover over **Clean 4**, click **+**, select **Output** (download/save icon), and verify the **Output** node is connected directly to **Clean 4**.

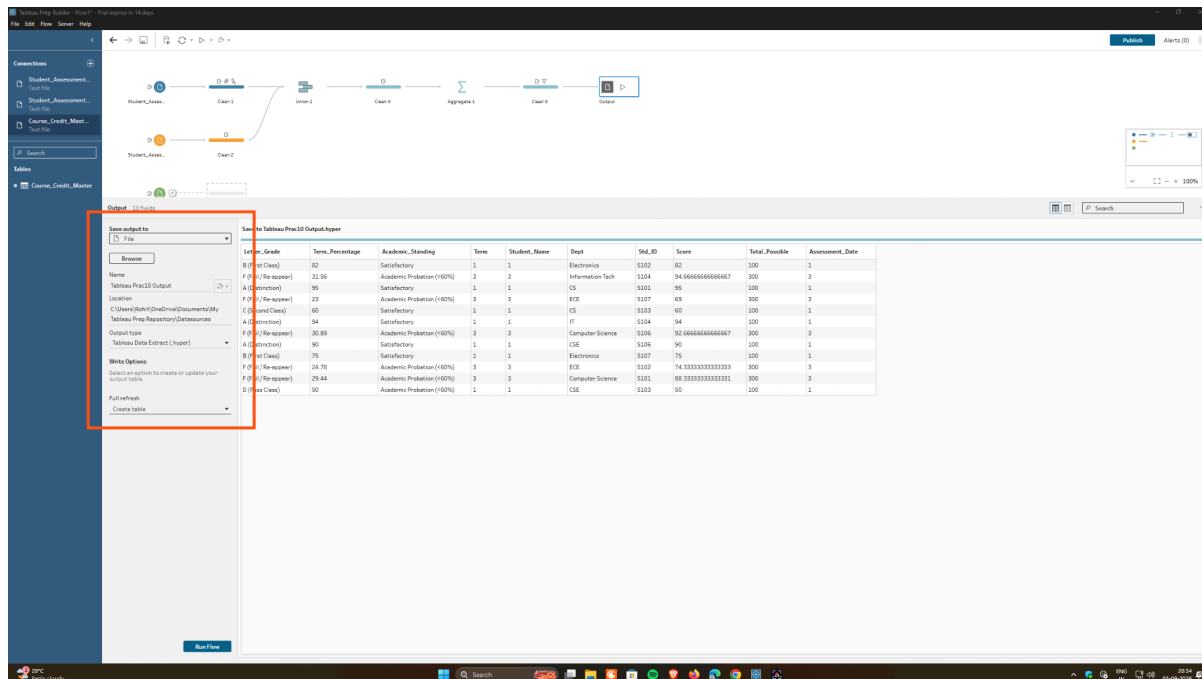
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The screenshot shows the Tableau Prep Builder interface. The 'Output' node is highlighted, and a context menu is open. The 'Output' option is selected. The data table below shows the output of the 'Output' node.

Letter_Grade	Term_Percentage	Academic_Standings	Term	Student_Name	Dept	Std_ID	Score	Total_Possible	Assessment_Date
A (Distinction)	82	Satisfactory	1	1	Electronics	S102	82	100	1
B (First Class)	74.78	Academic Probation (<60%)	3	3	Information Tech	S104	94.66666666666667	300	3
C (Second Class)	60	Satisfactory	1	1	CS	S101	90	100	1
D (Pass Class)	50	Academic Probation (<60%)	3	3	ECE	S107	69	300	3
F (Fail / Re-assess)	23	Satisfactory	1	1	CS	S103	60	100	1
A (Distinction)	96	Satisfactory	1	1	IT	S104	94	100	1
B (First Class)	75	Satisfactory	1	1	Computer Science	S106	90	100	1
C (Second Class)	60	Satisfactory	1	1	Electronics	S107	75	100	1
D (Pass Class)	50	Academic Probation (<60%)	3	3	ECE	S102	74.33333333333333	300	3
F (Fail / Re-assess)	29.44	Academic Probation (<60%)	3	3	Computer Science	S101	88.33333333333333	300	3
A (Distinction)	94	Academic Probation (<60%)	1	1	CSE	S103	90	100	1

Step 2: Select the **Output** node and configure **Save to File**, choose **.hyper** or **.csv**, set the desired **Location**, name the file **Student_Performance_Master_Cleaned**, and select **Create table** as the write option.



The screenshot shows the Tableau Prep Builder interface. The 'Output' node is highlighted, and the 'Save to File' dialog is open. The 'Save to File' dialog shows the 'Save to File' option selected, and the 'Save to File' dialog is open. The 'Save to File' dialog shows the 'Save to File' option selected, and the 'Save to File' dialog is open.

Letter_Grade	Term_Percentage	Academic_Standings	Term	Student_Name	Dept	Std_ID	Score	Total_Possible	Assessment_Date
B (First Class)	82	Satisfactory	1	1	Electronics	S102	82	100	1
F (Fail / Re-assess)	21.55	Academic Probation (<60%)	3	3	Information Tech	S104	94.66666666666667	300	3
A (Distinction)	96	Satisfactory	1	1	CS	S101	90	100	1
F (Fail / Re-assess)	23	Academic Probation (<60%)	3	3	ECE	S107	69	300	3
C (Second Class)	60	Satisfactory	1	1	CS	S103	60	100	1
A (Distinction)	94	Satisfactory	1	1	IT	S104	94	100	1
B (First Class)	75	Satisfactory	1	1	Computer Science	S106	90	100	1
C (Second Class)	60	Satisfactory	1	1	Electronics	S107	75	100	1
D (Pass Class)	50	Academic Probation (<60%)	3	3	ECE	S102	74.33333333333333	300	3
F (Fail / Re-assess)	29.44	Academic Probation (<60%)	3	3	Computer Science	S101	88.33333333333333	300	3
A (Distinction)	94	Academic Probation (<60%)	1	1	CSE	S103	90	100	1

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Step 3: Click **Run Flow** (blue button) or the ► icon on the Output node to process the entire pipeline, then verify the **"The flow ran successfully in X seconds."** notification appears.

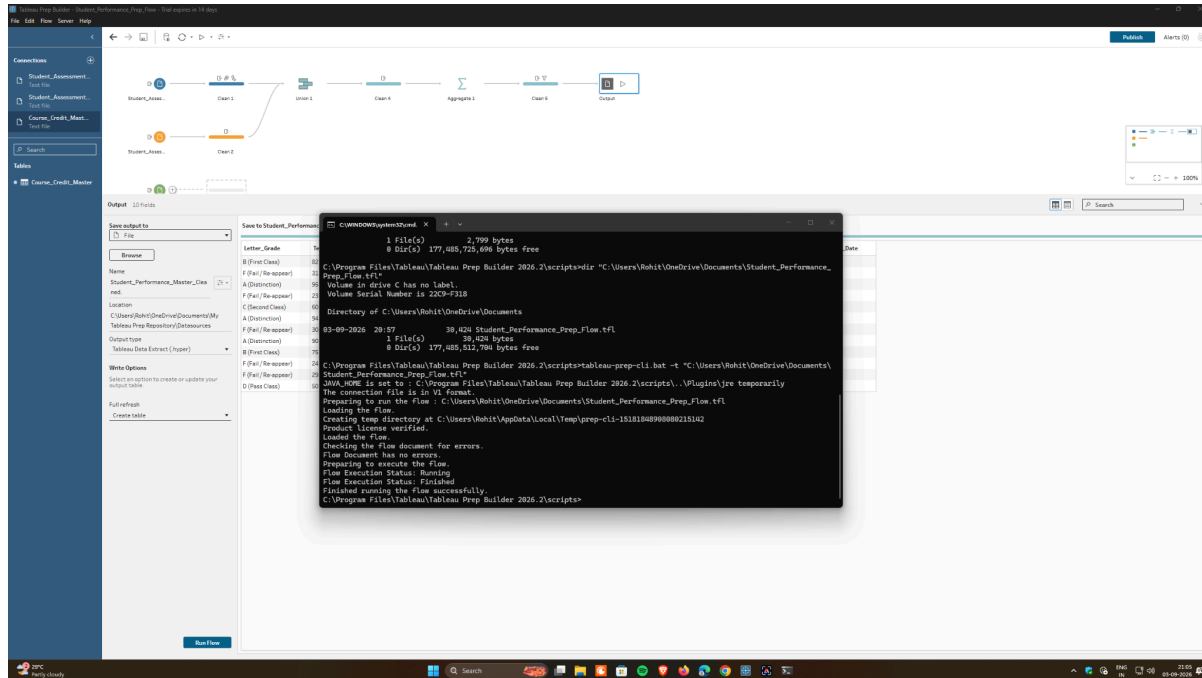
The screenshot shows the Tableau Prep Builder interface. The top pane displays a flow diagram with nodes for 'Student_Class', 'Exam 1', 'Exam 2', 'Aggregate 1', 'Exam 3', and 'Output'. The bottom pane shows the output data table, which has been saved to 'Student_Performance_Master_Cleaned.hyper'. A notification box in the center states 'Finished Running Flow' with a green checkmark and 'Total time 00:00'. The output table contains the following data:

Letter_Grade	Term_Percentage	Academic_Standings	Score	Student_Rank	Dept	Std_ID	Score	Total_Possible	Assessment_Date	
B (First Class)	82	Satisfactory	1	1	Electronics	S102	82	100	1	
F (Fail / Re-assess)	21.56	Academic Probation (<60%)	3	3	Information Tech	S104	94.66666666666667	300	3	
A (Distinction)	95	Satisfactory	1	1				100	1	
F (Fail / Re-assess)	23	Academic Probation (<60%)	3	3				300	3	
C (Second Class)	60	Satisfactory	1	1				100	1	
A (Distinction)	94	Satisfactory	1	1				100	1	
F (Fail / Re-assess)	30.89	Academic Probation (<60%)	3	3				300	3	
A (Distinction)	90	Satisfactory	1	1				100	1	
B (First Class)	75	Satisfactory	1	1				100	1	
F (Fail / Re-assess)	24.78	Academic Probation (<60%)	3	3				333333333	300	3
F (Fail / Re-assess)	29.44	Academic Probation (<60%)	3	3				333333333	300	3
D (Pass Class)	50	Academic Probation (<60%)	1	1				100	1	

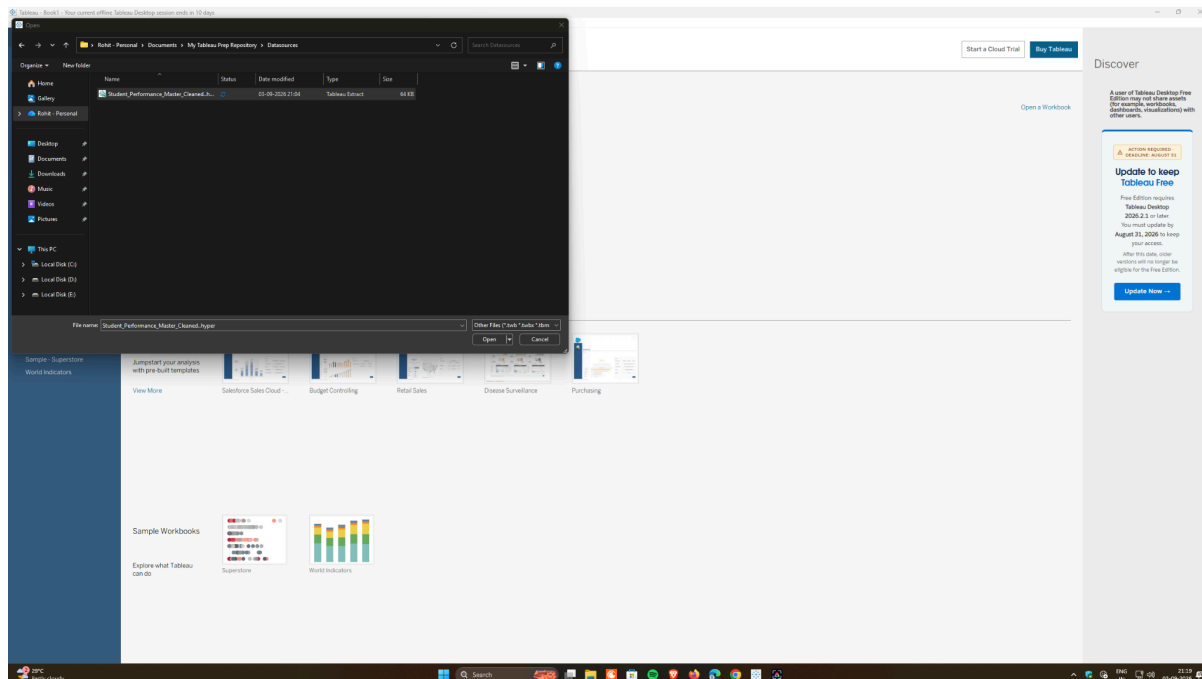
Step 4: Go to **File** → **Save As**, save the workflow as **Student_Performance_Prep_Flow.tfl**, and optionally automate it using **Tableau Prep CLI** or **Tableau Prep Conductor** with a scheduled refresh.

The screenshot shows the Tableau Prep Builder interface with the 'Save As' dialog box open. The dialog box displays the file name 'Student_Performance_Prep_Flow.tfl' and the save type 'Tableau Prep Flow (.tfl)'. The background shows the same flow diagram and output table as the previous screenshot.

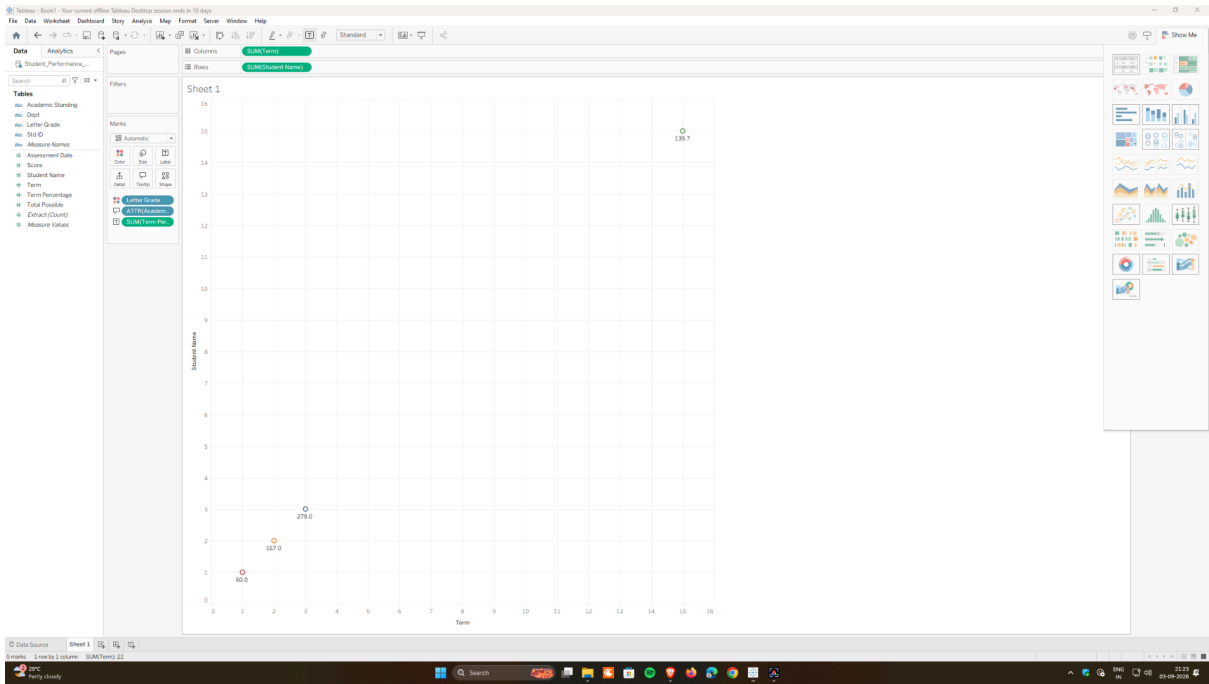
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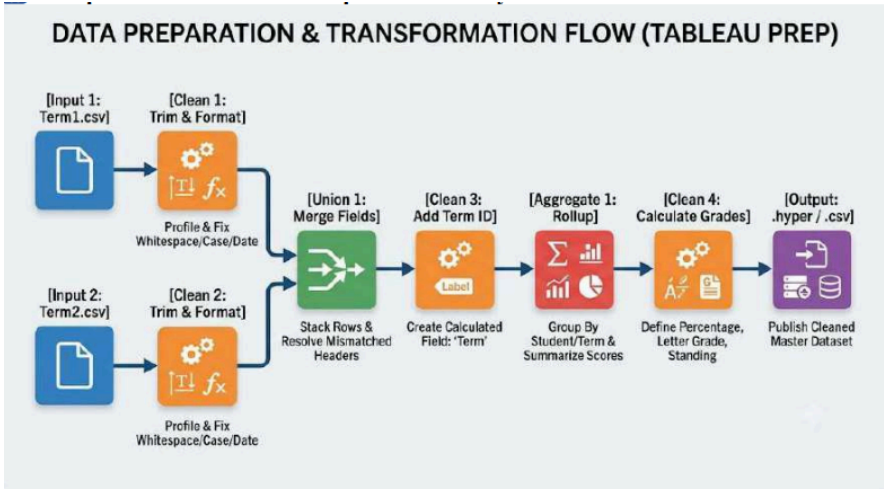
Step 5: Open **Tableau Desktop**, connect to **Student_Performance_Master_Cleaned.csv/.hyper**, then create the dashboard by placing **Student_Name** → **Rows**, **Term** → **Columns**, **Term_Percentage** → **Text**, **Letter_Grade** → **Color**, and **Academic_Standing** → **Tooltip**.



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Complete End-to-End Flow Pipeline Summary



Pipeline Stage	Step Type	Key Actions Performed	Output Result
Input Layer	Data Ingestion	Connect to Term1.csv, Term2.csv, and Course_Master.csv	Raw tables loaded on canvas
Cleaning Layer	Clean Steps 1 & 2	Trim whitespace, standardize dates, uppercase names	Structured singelsource tables

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Standardization	Group Values	Consolidate aliases (CS, CSE, Comp Sci\$\rightarrow\$ Computer Science)	Clean department dimensions
Integration Layer	Union Step	Stack Term 1 & 2; drag-and-drop mismatched headers	Unified schema (30 assessment rows)
Aggregation Layer	Aggregate Step	Group by student & term; compute Sum(Marks) and Avg(Score)	14 summary records
Business Logic	Clean Step 4	Filter dropped students; calculate %, Letter_Grade, and Standing	Analytical scorecard metrics
Export Layer	Output Step	Export to Student_Performance_Master_Cleaned .hyper / .csv	Production-ready dashboard source